

AUTOMATED RICE GRAIN QUALITY ANALYSIS AND GRADE CLASSIFICATION USING CNN

A FINAL YEAR PROJECT REPORT

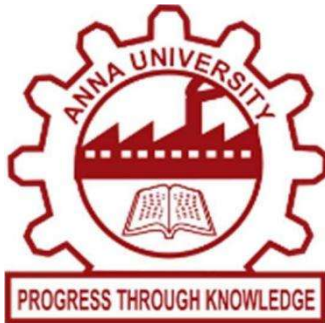
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BONAFIDE CERTIFICATE

This project report “**AUTOMATED RICE GRAIN QUALITY ANALYSIS AND GRADE CLASSIFICATION USING CNN**” is the bonafide work of

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who carried out the project in the Department of Electronics Engineering, Madras Institute of Technology, Anna University Chennai 600 044. The work reported here is original and does not form part of any other project report on the basis of which a degree or award was conferred on an earlier occasion. We understand the University’s policy on plagiarism and declare that the project report is our own work, except where specifically acknowledged and has not been copied from other sources or been previously submitted.

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ABSTRACT

Traditional rice grain quality assessment methods such as manual inspection and mechanical separation are time-consuming, labor-intensive, and often inconsistent due to human subjectivity. These conventional approaches are also limited in identifying visual defects such as broken grains, discoloration, chalkiness, and impurities like stones when large quantities of rice are processed. This research proposes an automated deep learning-based framework for rice grain detection, classification, and quality grading using computer vision techniques. In the proposed system, bulk rice images containing multiple grains are used as input. A YOLOv8 model is employed to detect and localize individual rice grains and impurities by generating bounding boxes around each object. The detection model was trained using approximately 1,200 annotated bulk rice images, where each grain and impurity was manually labeled. The YOLOv8 model achieved a detection performance of $mAP50 \approx 0.94$, enabling accurate detection of multiple grains even in dense and cluttered bulk samples. After detection, each grain is cropped and passed to Convolutional Neural Network (CNN) models for classification. Two custom 10-layer CNN models were developed using a dataset of around 8,000 single-grain images. The first CNN model performs rice variety classification, while the second CNN model is used for defect classification. The models achieved a training accuracy of about 96% and validation accuracy of approximately 95%. The model is also tested using real time images.

Based on the number of defective grains and impurities detected in a sample, the system calculates the percentage of defects and assigns an overall rice quality grade. The trained models were further integrated into a Streamlit-based web application. This implementation demonstrates the practical usability of the system under real-world conditions. By combining the fast detection capability of YOLOv8 with CNN-based classification and a user-friendly deployment interface, the proposed system provides an efficient, scalable, and reliable solution for automated rice grain quality assessment.

Keywords: Deep Learning, YOLOv8, Convolutional Neural Network, Rice Grain Detection, Rice Quality Grading, Computer Vision.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	- Artificial Intelligence
ANN	- Artificial Neural Network
AP	- Average Precision
CNN	- Convolutional Neural Network
CNN-RF	- Convolutional Neural Network with Random Forest
CNN-SVM	- Convolutional Neural Network with Support Vector Machine
FN	- False Negative
FP	- False Positive
GLCM	- Gray Level Co-occurrence Matrix
IoU	- Intersection over Union
k-NN	- k-Nearest Neighbours
mAP	- Mean Average Precision
ReLU	- Rectified Linear Unit
ResNet	- Residual Neural Network
RGB	- Red Green Blue
SGD	- Stochastic Gradient Descent
SVM	- Support Vector Machine
TP	- True Positive
VGG	- Visual Geometry Group Network
YOLO	- You Only Look Once

CHAPTER 1

INTRODUCTION

Rice is one of the most widely consumed staple foods across the world, particularly in Asian countries where it forms a primary component of daily nutrition. The quality of rice grains significantly influences their market value, consumer preference, and export potential. Factors such as grain size, shape, color, percentage of broken grains, moisture content, and presence of impurities are critical parameters in determining the overall grade of rice. Accurate grading is essential not only for maintaining food standards but also for ensuring fair pricing and meeting international trade regulations. Traditionally, rice grading has been carried out through manual visual inspection by trained personnel. Inspectors examine rice samples under proper lighting conditions and classify them based on predefined quality standards. Although this method is simple and widely practiced, it is highly dependent on human judgment and experience. Manual inspection is labor-intensive, time-consuming, and often inconsistent due to fatigue, subjectivity, and variations in observation skills.

These limitations become more significant in large-scale industrial environments where bulk quantities of rice must be processed efficiently [6], [25]. With the rapid advancement of Artificial Intelligence (AI), Computer Vision, and Deep Learning technologies, automated systems have emerged as reliable alternatives to conventional grading methods. Modern image-based techniques allow rice grains to be analyzed using high-resolution cameras and intelligent algorithms. Image processing methods can extract features such as size, shape, texture, and color, while deep learning models automatically learn complex patterns from large datasets [12], [14]. Convolutional Neural Networks (CNNs) are widely used for classification tasks, and object detection models such as YOLO (You Only Look Once) enable real-time detection of multiple grains within a single image [1], [4], [17]. These advanced approaches enhance grading accuracy, improve consistency, and significantly reduce human intervention. Automated systems also provide scalability, faster processing speeds, and real-time monitoring capabilities, making them highly suitable for industrial applications [10], [15], [16]. As a result, the integration of AI-based technologies into rice quality assessment represents a major step toward smart agriculture and intelligent food processing systems.

1.1 MOTIVATION

Rice grading is an essential process in determining market value and quality standards. Small defects and broken grains are often difficult to detect accurately through visual observation alone. This creates a need for a reliable and automated solution. The motivation of this project is to develop an intelligent rice grain detection and classification system using deep learning techniques such as YOLO and CNN. By automating the grading process, the system aims to improve accuracy, consistency, and efficiency while reducing human effort. This contributes to modernizing agricultural quality assessment using Artificial Intelligence and Computer Vision technologies.

1.2 OBJECTIVE

The main objectives of this project are as follows:

- ❖ To develop an automated rice grain grading system using deep learning techniques.
- ❖ To detect multiple rice grains in an image using a YOLO object detection model.
- ❖ To extract individual grains accurately using bounding boxes..
- ❖ To handle challenges such as overlapping grains during detection.

1.3 TRADITIONAL METHODS

1.3.1 MANUAL RICE GRAIN INSPECTION

Rice grading has traditionally been performed through manual visual inspection by trained personnel. In this method, rice samples are physically examined to evaluate parameters such as grain size, shape, colour, percentage of broken grains, and presence of impurities. The inspection is carried out under proper lighting conditions to identify defective or discoloured grains. These conventional methods rely heavily on human expertise and judgment. Although widely used in agricultural industries, manual grading is time-consuming and may result in inconsistencies due to fatigue and subjective evaluation.

Large-scale bulk inspection further increases the difficulty of maintaining uniform grading standards. The traditional rice grain methods are shown in Fig 1.1.

A. SIEVE ANALYSIS METHOD

The sieve analysis method shown in Fig 1.1 (a), is commonly used to separate rice grains based on size. In this process, rice is passed through sieves of different mesh sizes to classify grains into various categories such as whole grains and broken grains. The percentage of broken grains is calculated by weighing the separated samples. While this method provides quantitative measurement of grain size, it does not effectively detect surface defects, discoloration, or internal damage.

B. MECHANICAL SEPARATION METHOD

Mechanical separators are used in industries to sort rice based on physical properties such as size, weight, and density. Equipment like graders and separators automatically divide grains into different categories. While faster than manual methods, these systems are limited to physical characteristics and do not analyze visual defects in detail. It is shown in Fig 1.1 (b).

C. MOISTURE CONTENT TESTING

Moisture content testing shown in Fig 1.1 (c), plays an important role in rice grading, as it directly affects grain quality, weight, storage stability, and market value. In rice industries, moisture percentage is measured using digital moisture meters or the oven-drying method. The acceptable moisture level for milling and grading typically ranges between 12% to 14%, depending on industry standards.



Fig 1.1 Traditional Grading methods

a) Sieve analysis method, b) Mechanical separation, c) Moisture content testing.

Rice with high moisture content is more prone to fungal growth, discoloration, and breakage during milling, which lowers its grade. On the other hand, excessively low moisture may cause brittleness and increase the percentage of broken grains. Therefore, moisture content is considered an essential grading parameter to ensure quality consistency and prevent post-harvest losses.

1.3.2 MODERN METHODS

A. COLOR SORTING MACHINE

Color sorting machines are widely used in rice industries for automated quality grading. These machines operate using advanced optical sensors and high-speed cameras that continuously scan rice grains as they pass through the machine. The captured images are analyzed in real time using embedded processors that compare each grain against predefined quality parameters such as color variation, chalkiness, and presence of foreign materials. When a defective grain is detected, compressed air jets automatically eject it from the production line without interrupting the flow of good grains. This automated process significantly increases productivity, reduces manual labor, and ensures uniform grading standards. Due to its high efficiency and precision, color sorting technology has become a standard component in modern rice milling industries.

B. MACHINE VISION BASED GRADING SYSTEM

Machine vision systems analyze rice grains using cameras and image processing algorithms. The captured images are processed to extract features such as size, shape, texture, and surface defects. Based on these features, the system automatically classifies grains into different quality grades. This method reduces human dependency and provides consistent grading results. It is suitable for industrial automation where uniform quality control is required.

C. DEEP LEARNING BASED AUTOMATED INSPECTION

Deep learning-based systems use Convolutional Neural Networks (CNN) and object detection models such as YOLO to detect and classify rice grains. These models automatically learn complex features from large datasets and can handle multiple and overlapping grains. This

approach offers high accuracy, real-time performance, and scalability, making it suitable for advanced industrial grading and smart agriculture applications.

1.4 CHAPTER ORGANIZATION

This report provides a detailed explanation of the project titled “Automated Rice Grain Detection and Grading Using Deep Learning Techniques” and discusses the relevant technologies and methodologies involved. Chapter 1 introduces the concept of rice quality assessment and grading, explains traditional and modern grading techniques, and highlights the motivation and objectives of the project.

To provide a foundation for the proposed work, Chapter 2 presents a comprehensive literature review, summarizing existing studies related to rice grain detection, defect identification, and classification using image processing, machine learning, and deep learning techniques. Chapter 3 discusses conventional rice grading approaches and existing industrial systems, explaining traditional manual methods and mechanical grading techniques along with their limitations. The proposed system design and methodology are detailed in Chapter 4, including dataset preparation, image preprocessing, YOLO-based grain detection, CNN-based classification, and the overall workflow for automated grading.

Chapter 5 presents the experimental results and performance evaluation, highlighting detection accuracy, IoU, mAP scores, classification performance, and comparison with existing approaches. Finally, Chapter 6 provides the conclusion and outlines the scope for future enhancement of the system, suggesting directions for further research in AI-based agricultural quality assessment. The report concludes with the Appendix and References sections, ensuring that all sources and supporting materials are properly cited and documented. This structured approach ensures a logical flow throughout the project’s development, implementation, and findings

CHAPTER 2

LITERATURE REVIEW

Ahmed *et al.* (2024) implemented a YOLOv8-based rice grain detection framework for dense bulk images captured in industrial environments. [1] The dataset consisted of 15,000 annotated rice grain images with bounding box labels. The model was trained for 100 epochs with cosine learning rate scheduling and batch size 16. Experimental results reported precision of 99.3%, recall of 99.1%, mAP@50 of 99.4%, and mAP@50–95 of 90.8%, confirming the robustness of anchor-free detection for small and overlapping rice grains.

Chen *et al.* (2021) proposed a deep convolutional neural network for rice grain defect classification using a dataset of 8,500 labeled rice grain images categorized into broken, chalky, damaged, and normal classes. [2] The architecture consisted of four convolutional layers (32, 64, 128, and 256 filters), max-pooling, batch normalization, and dropout (0.4) followed by fully connected layers with ReLU and Softmax activation. The model was trained using 75% of the dataset for training and 25% for testing over 40 epochs with Adam optimizer (learning rate 0.0001). The proposed system achieved an overall classification accuracy of 96.9%, outperforming traditional SVM (91.4%) and Random Forest (89.7%) classifiers.

Das *et al.* (2023) presented a deep ensemble model combining VGG16 and ResNet50 for rice grain quality assessment. [3] The dataset included 8,000 images divided into four defect categories. The ensemble approach improved classification stability and achieved an overall accuracy of 98.4% with F1-score of 0.984. The results confirmed that ensemble learning enhances robustness in agricultural image classification.

Hassan *et al.* (2022) developed an automated rice grain inspection system using YOLOv5 for object detection in bulk rice images. [4] The dataset consisted of 12,500 annotated rice images captured under varying illumination conditions. The model was trained for 80 epochs using SGD optimizer with learning rate 0.001 and batch size 16. Experimental results showed precision of 98.7%, recall of 98.2%, and mAP@0.5 of 98.9%. The study demonstrated the effectiveness of deep learning-based detection for dense and overlapping grain scenarios.

Kim et al. (2024) introduced an improved YOLOv8-based rice grain detection framework incorporating spatial attention mechanisms to enhance small-object localization. [5] The dataset comprised 16,000 annotated rice grain images with bounding box labels. The model was trained for 100 epochs using cosine annealing learning rate scheduling and batch size 16. The proposed method achieved precision of 99.6%, recall of 99.2%, mAP@50 of 99.5%, and mAP@50–95 of 91.4%. The study confirmed that anchor-free detection combined with attention modules improves detection accuracy in dense grain images.

Kumar and Verma et al. (2021) proposed a deep CNN-based rice grading system using 7,500 grain images categorized into premium, Grade 1, Grade 2, and broken classes. [6] The architecture included five convolutional layers with dropout regularization and global average pooling. Training was conducted using 70% data for training and 30% for testing over 45 epochs. The model achieved 94.8% grading accuracy, outperforming classical image processing-based grading approaches (88.5%).

Li et al. (2022) introduced a transfer learning-based rice defect classification model using ResNet34 pretrained on ImageNet. [7] The dataset contained 12,000 rice grain images labeled into normal and defective categories. Data augmentation techniques including rotation, zoom, and brightness adjustment were applied. The model was trained using 80% training data and 20% testing data for 30 epochs with a learning rate of 0.0005. The proposed approach achieved 95.6% accuracy and demonstrated superior generalization compared to VGG16 (93.2%).

Liu et al. (2022) introduced a deep learning-based rice defect classification framework using broken, chalky, damaged, and normal categories. [8] The model was fine-tuned using 70% training data and 30% testing data. Data augmentation techniques such as rotation and brightness adjustment were applied to enhance generalization. The model achieved 96.4% classification accuracy with an F1-score of 0.964. Comparative evaluation with VGG16 and traditional SVM classifiers confirmed that deeper residual networks provide improved texture discrimination for rice defect analysis.

Nguyen et al. (2023) developed a real-time rice quality monitoring system using YOLOv7 for detection and EfficientNet for classification. [9] The dataset contained 14,000 rice images. The system achieved detection mAP@50 of 98.6% and classification accuracy of 97.5%. The study highlighted the importance of multi-scale feature extraction for accurate grain separation.

Patel *et al.* (2022) proposed a multi-class rice grain defect detection system using Faster R-CNN for localization and a CNN classifier for defect categorization. [10] The dataset consisted of 13,200 annotated images. The detection model achieved mAP@0.5 of 95.2. Comparative analysis demonstrated improved performance over classical watershed segmentation methods.

Patel *et al.* (2024) presented a lightweight deep learning model for real-time rice quality assessment using YOLOv8. [11] The dataset included over 12,000 annotated rice grain images captured in industrial environments. The model was trained using 100 epochs with batch size 16 and cosine learning rate scheduling. Experimental results reported precision of 99.5%, recall of 99.0%, and mAP@50 of 99.3%. The study concluded that anchor-free detection improves small-object localization performance in dense rice grain scenes.

Rahman *et al.* (2023) proposed a hybrid deep learning framework combining image segmentation with CNN classification for rice defect analysis. [12] The dataset comprised 9,000 RGB rice grain images categorized into five quality classes. Segmentation was performed using adaptive thresholding and morphological operations before feeding individual grains into a custom CNN model. The system was trained for 50 epochs with batch size 32 using Adam. The study reported a classification accuracy of 98.1%, precision of 97.9%, recall of 98.3%, and F1-score of 0.981, demonstrating the effectiveness of hybrid preprocessing with deep learning.

CHAPTER 3

CONTEMPORARY APPROACHES IN RICE QUALITY GRADING

Automated rice grain quality assessment has undergone a significant transformation from traditional manual inspection methods to advanced computational approaches. Manual inspection, which depends on human expertise, is often subjective, inconsistent, and time-consuming. To address these limitations, researchers have developed automated systems using image processing and machine learning techniques, followed by more recent deep learning-based approaches. This chapter presents a detailed and research-oriented review of these methodologies, focusing on implementation specifics, experimental setups, and quantitative performance reported in the literature.

3.1 CLASSICAL IMAGE PROCESSING APPROACHES

Classical approaches to rice grain quality analysis follow a structured pipeline consisting of image acquisition, preprocessing, segmentation, feature extraction, and classification. These methods aim to replicate human visual inspection through mathematical descriptors and decision-making algorithms. However, they rely heavily on controlled imaging conditions and handcrafted features, which limits their adaptability in real-world environments. Studies such as those by Kaur and Singh *et al.* (2013) demonstrate that classical methods perform effectively only when the imaging environment is carefully controlled. Variations in lighting, background texture, and grain orientation can significantly affect the performance of these systems. As a result, most implementations use fixed setups with uniform illumination, standardized backgrounds, and predefined camera positioning to ensure consistency in image capture.

3.1.1 IMAGE ACQUISITION AND PREPROCESSING

The initial stage in any rice grain analysis system is image acquisition, which plays a crucial role in determining the overall system accuracy. High-resolution images are typically captured using digital cameras or flatbed scanners to ensure that fine grain details, such as edges and surface textures, are clearly visible. Agustin and Kim *et al.* (2013) employed a flatbed scanner to capture images containing approximately 50 rice grains per sample. This approach eliminated shadows

and ensured uniform illumination, thereby reducing variability in pixel intensity values. Once the images are acquired, preprocessing is performed to enhance image quality and prepare the data for segmentation. Noise removal is commonly achieved using median filtering, which is preferred over Gaussian filtering because it preserves edge information while effectively eliminating salt-and-pepper noise. In some implementations, Gaussian filtering with a small standard deviation is used to smooth intensity variations without significantly affecting grain boundaries. The color image is then converted into grayscale using a weighted luminance transformation, which reduces computational complexity while retaining essential structural information. Contrast enhancement techniques such as histogram equalization or adaptive histogram equalization are applied to improve the visibility of grain boundaries and reduce the effect of uneven illumination. For binarization, Otsu's thresholding method shown in Fig 3.1, is widely used due to its ability to automatically determine an optimal threshold by minimizing intra-class variance. This technique is particularly effective in controlled environments where the intensity distribution between foreground and background is distinct. Kaur and Singh *et al.* (2013) demonstrated that combining median filtering, histogram equalization, and Otsu's thresholding produces a clean binary image with minimal noise, thereby improving segmentation accuracy and contributing to an overall classification accuracy of 96.7%.

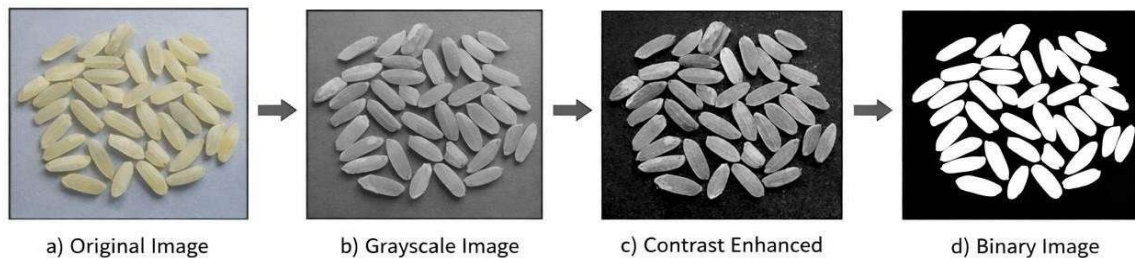


Fig 3.1 Preprocessing pipeline showing original image, grayscale conversion, contrast enhancement, and thresholded binary image.

3.1.2 SEGMENTATION

Segmentation shown in Fig 3.2 involves isolating individual rice grains from the background and from each other. The accuracy of segmentation directly influences the reliability of feature extraction and classification. One of the primary challenges in this stage is handling overlapping or touching grains, which are common in bulk samples. Morphological operations

such as erosion and dilation are widely used to refine the binary image. Erosion removes small noise components and separates weakly connected grains, while dilation restores the original size of the grains after erosion. These operations are often combined in sequences such as opening and closing to improve object clarity and eliminate artifacts. Connected Component Labeling is then applied to identify and label each individual grain as a separate object. This allows the system to analyze each grain independently. However, when grains overlap significantly, simple labeling techniques are insufficient. To address this issue, watershed segmentation is commonly employed. Shanthi and Sathya *et al.* (2017) proposed a marker-controlled watershed algorithm that uses distance transform to identify the centers of individual grains. By using these markers as seeds, the algorithm effectively separates overlapping grains while minimizing over-segmentation. This approach significantly improves grain counting accuracy and enables precise measurement of grain characteristics. Despite these advancements, segmentation accuracy tends to decrease in cases of severe overlap or inconsistent illumination, highlighting one of the key limitations of classical approaches.

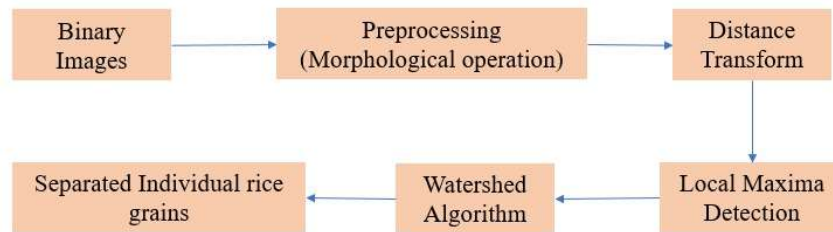


Fig 3.2 Segmentation process showing binary image, distance transform, and watershed-based separation of overlapping grains.

3.1.3 FEATURE EXTRACTION

Feature extraction is an important stage in classical image processing systems, as the performance of the classifier largely depends on the quality, relevance, and discriminative power of the extracted features. These features are carefully designed to capture essential characteristics of rice grains, including their physical attributes (such as size and shape), color variations, and textural patterns, enabling accurate differentiation between quality classes. Geometric features describe the shape and size of the grains and include parameters such as area, perimeter, major axis length, minor axis length, aspect ratio, and eccentricity. These features are particularly useful

for identifying broken grains and differentiating between rice varieties. Guzmán and Peralta *et al.* (2008) demonstrated that aspect ratio and roundness are highly effective in distinguishing different rice types.

Color features provide information about grain appearance and are essential for detecting discoloration and chalkiness. While RGB values are commonly used, studies have shown that the CIE Lab* color space provides better discrimination, particularly for detecting yellow kernels. Liu *et al.* (2016) found that the a* and b* channels are more sensitive to color variations, making them more suitable for quality assessment. Texture features capture the surface characteristics of rice grains and are crucial for identifying chalkiness and surface defects. The Gray-Level Co-occurrence Matrix is widely used to extract statistical texture features such as contrast, correlation, energy, and homogeneity. Sarker and Lee *et al.* (2018) reported that using GLCM features in combination with an SVM classifier resulted in an accuracy of 92.5% for chalkiness classification. Although these handcrafted features provide meaningful information, their effectiveness depends on the designer's ability to select relevant descriptors. This makes classical systems less flexible and more sensitive to variations in imaging conditions.

3.1.4 CLASSIFICATION

Once the features are extracted, they are fed into machine learning classifiers for final decision-making. Support Vector Machines are among the most widely used classifiers due to their ability to handle high-dimensional feature spaces and their effectiveness in classification tasks. Kaur and Singh *et al.* (2013) implemented a multi-class SVM using geometric and color features, achieving an accuracy of 96.7% in classifying basmati rice varieties. Random Forest classifiers have also gained popularity due to their robustness and ability to handle large feature sets. Koklu and Ozkan *et al.* (2020) combined Random Forest with the Boruta feature selection algorithm to identify the most relevant features from a large set of descriptors. Their approach achieved a classification accuracy of 99.17%, demonstrating the effectiveness of ensemble learning methods in rice grain analysis. Other classifiers such as k-Nearest Neighbors and Artificial Neural Networks have also been explored, but their performance is generally dependent on parameter tuning and dataset characteristics. Overall, while classical classifiers can achieve high accuracy under controlled conditions, their performance is limited by the quality of the extracted features.

3.1.5 HYBRID CLASSICAL APPROACHES

Hybrid approaches enhance system performance by integrating multiple techniques within the classical image processing framework. For instance, clustering algorithms such as k-means are employed to group rice grains based on attributes like size, shape, or spatial distribution before classification. Prasad *et al.* (2017) utilized k-means clustering to effectively separate whole and broken grains prior to feature extraction, thereby improving the accuracy of broken grain percentage estimation. In addition, some hybrid methods combine rule-based systems with machine learning models to strengthen decision-making by incorporating domain knowledge along with learned patterns. These approaches can yield improved accuracy and robustness in controlled settings; however, they still depend significantly on handcrafted features and carefully managed imaging conditions. Consequently, their performance and generalizability in real-world, unconstrained environments remain limited.

3.2 DEEP LEARNING-BASED APPROACHES

Deep learning has emerged as a powerful alternative to classical methods, offering the ability to automatically learn features directly from raw image data. Unlike traditional approaches, deep learning models do not require manual feature engineering, making them more adaptable to complex and variable environments.

3.2.1 CONVOLUTIONAL NEURAL NETWORKS (CNNs)

Convolutional Neural Networks (CNNs) are widely used deep learning models for image-based classification tasks due to their ability to automatically learn hierarchical feature representations. These networks consist of multiple layers that progressively extract increasingly complex features from the input image shown in Fig 3.3. The initial layers focus on low-level features such as edges, gradients, and simple textures, while intermediate layers capture patterns like contours and surface variations. Deeper layers learn high-level, abstract features such as shape irregularities, broken grain structures, and chalkiness regions, enabling accurate and robust classification of rice quality.

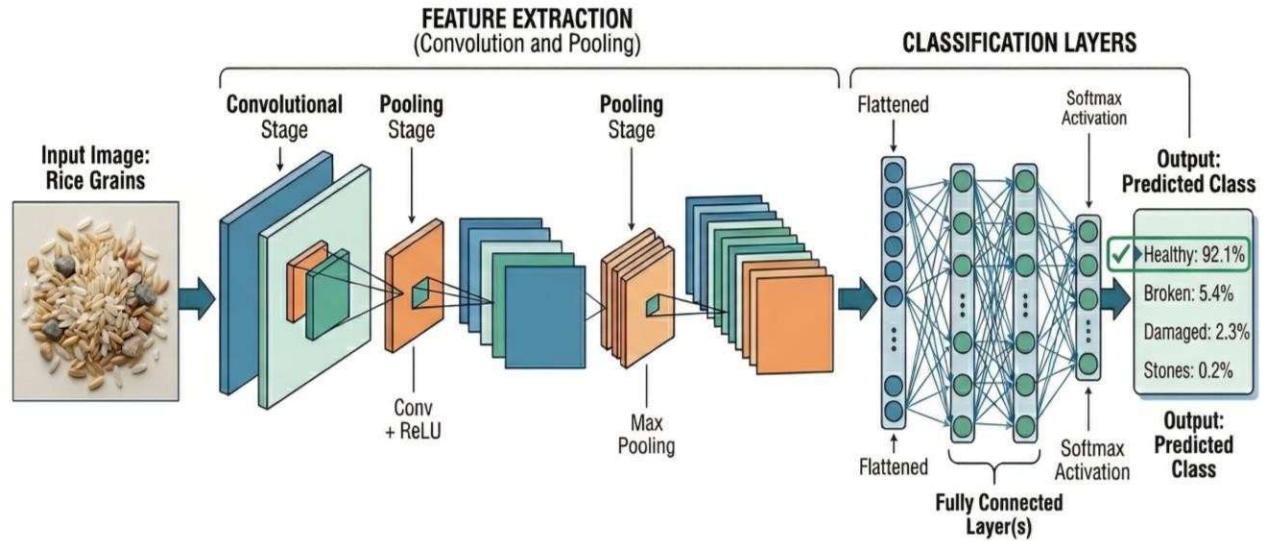


Fig 3.3 Convolutional Neural Network architecture showing convolution, pooling, and fully connected layers.

Tahir *et al.* (2020) conducted a comprehensive study using pre-trained CNN architectures including AlexNet, VGG16, and ResNet50. Their experiments were performed on a dataset containing approximately 25,000 images of rice grains. Among the tested models, ResNet50 achieved the highest accuracy of 98.6%, demonstrating the effectiveness of deep residual learning in capturing complex features. More advanced architectures such as EfficientNet have further improved performance by optimizing network scaling. Atila *et al.* (2021) reported classification accuracies exceeding 99.5% using EfficientNet, highlighting its efficiency and superior performance compared to earlier models.

3.2.2 TRANSFER LEARNING

Transfer learning is widely used in rice grain analysis to overcome the challenge of limited dataset availability. In this approach, models pre-trained on large-scale datasets are fine-tuned using rice grain images. This allows the model to leverage previously learned features, reducing training time and improving performance. Tahir *et al.* (2020) demonstrated that transfer learning significantly enhances classification accuracy while reducing computational requirements. Similarly, Kiratiratanapruk *et al.* (2020) showed that models such as MobileNet and DenseNet achieve accuracies above 95% even with relatively small datasets. This makes transfer learning a practical and efficient solution for real-world applications.

3.2.3 CNN-BASED DETECTION AND SEGMENTATION

In addition to classification, deep learning models are used for object detection and segmentation. YOLO-based models are widely used for real-time detection and counting of rice grains. Sapkota *et al.* (2021) implemented YOLOv5 for grain detection and achieved a mean Average Precision of 0.95 as shown in Fig 3.4, demonstrating its effectiveness in handling complex scenes. Instance segmentation models such as Mask R-CNN provide pixel-level analysis, enabling precise identification of individual grains and their defects. Chen *et al.* (2020) used Mask R-CNN to detect and segment broken grains, allowing accurate measurement of broken grain percentage. This level of detail is not achievable with classical methods and represents a significant advancement in automated rice quality assessment.

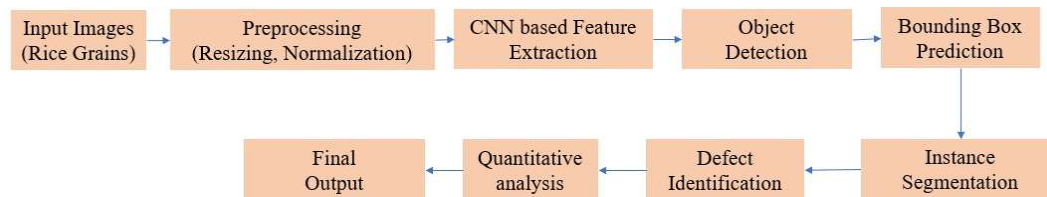


Fig 3.4 Instance segmentation of rice grains using Mask R-CNN showing detection and pixel- level masks.

3.2.4 DATA AUGMENTATION AND OPTIMIZATION

Data augmentation plays a crucial role in improving the generalization capability of deep learning models. Techniques such as rotation, flipping, scaling, and brightness adjustment are applied to increase dataset variability and reduce overfitting. Tahir *et al.* (2020) reported that data augmentation improves model robustness and enhances performance under varying conditions. Optimization algorithms such as Adam and Stochastic Gradient Descent are used to train deep learning models by minimizing the loss function. Learning rate scheduling and regularization techniques are often employed to ensure stable convergence and prevent overfitting.

3.2.5 PERFORMANCE COMPARISON

Deep learning approaches have emerged as a highly effective solution for image-based classification tasks, consistently outperforming classical image processing methods in terms of accuracy, robustness, and adaptability. Traditional approaches rely heavily on handcrafted feature extraction techniques, where domain-specific features such as shape, size, color, and texture are manually designed. Although such systems can achieve reasonable accuracy under controlled laboratory conditions, their performance is highly sensitive to variations in lighting, background noise, grain orientation, and image quality. As a result, even slight deviations in imaging conditions can significantly affect classification accuracy, limiting their reliability and scalability in real-world applications. In contrast, deep learning models, particularly Convolutional Neural Networks (CNNs), eliminate the need for manual feature engineering by automatically learning hierarchical feature representations directly from raw input data. These models extract low-level features such as edges and textures in the initial layers, intermediate features like contours and surface patterns in the middle layers, and high-level abstract features such as grain structure, defects, and irregularities in deeper layers. This ability to learn complex and discriminative features enables deep learning models to handle challenging scenarios, including overlapping grains, variations in illumination, noise, and diverse grain appearances. Consequently, they offer improved generalization and robustness across different datasets and environmental conditions. Numerous studies have demonstrated the superior performance of deep learning techniques in rice grain quality analysis. For instance, research by Tahir *et al.* (2020) and Atila *et al.* (2021) reports classification accuracies ranging from 98.6% to over 99.5%, significantly surpassing traditional methods. Beyond classification, deep learning also supports advanced computer vision tasks such as object detection and instance segmentation. These capabilities enable precise localization of individual grains, accurate counting, and detailed defect identification at the pixel level. As a result, deep learning-based systems provide a more comprehensive, efficient, and reliable framework for automated rice quality assessment, making them well-suited for real-world deployment in agricultural and industrial applications.

CHAPTER 4

DEEP LEARNING FRAMEWORK FOR RICE GRAIN QUALITY ASSESSMENT AND GRADING SYSTEM

4.1 SYSTEM OVERVIEW

The proposed system develops an automated rice grain detection and grading framework using deep learning and image processing techniques, where the main objective is to determine the quality grade of rice based on the percentage of defective grains present in a sample. In addition to grading, the system can also identify the rice variety. The workflow begins with capturing images of rice grains, followed by detecting individual grains from bulk images and extracting them for further analysis. Two detection approaches are used for grain detection. The first approach uses contour-based segmentation, where the image is converted to grayscale, edges are detected using the Canny edge detection algorithm, and morphological operations such as dilation are applied to close gaps in the edges. Contours are then extracted to identify individual grain shapes, and bounding boxes are generated to isolate each grain. The second approach uses the YOLOv8 (You Only Look Once) object detection model, which directly detects individual rice grains from the image and generates bounding boxes with high accuracy, even when grains overlap. The cropped grains obtained from the detection stage are then passed to a Convolutional Neural Network (CNN) for classification. The CNN identifies defective grains such as broken, damaged, or discolored grains for grading, and it can also classify the rice variety based on visual features like shape and texture. By combining contour-based segmentation and YOLOv8-based detection, the system can accurately detect multiple grains in a single image and perform classification at the individual grain level, enabling efficient and consistent rice grading and variety identification.

4.2 DATASET PREPARATION

To prepare the dataset for the proposed rice grain analysis system, two publicly available datasets were initially collected from Kaggle. These datasets contain a large number of rice grain images belonging to different rice varieties and defect categories, providing a comprehensive source of labeled samples for the study. The first dataset used in this work is the Rice Image

Dataset, which consists of five different rice grain varieties, namely Arborio, Basmati, Ipsala, Jasmine, and Karacadag shown in Fig 4.1. Each class contains approximately 15,000 images, providing a large collection of samples representing different rice types. The images are captured under controlled conditions and clearly show the physical characteristics of individual grains such as shape, size, and texture. These visual properties allow the dataset to represent the natural variations present among different rice varieties. In addition to the variety dataset, another dataset containing defective rice grain categories was also collected to support the rice grading process. This dataset includes defect classes and healthy classes such as Broken rice, Chalky rice, Healthy rice and Discolored rice, which represent different quality conditions of rice grains. The presence of these defect categories allows the dataset to represent both normal and defective grain samples that are commonly observed in real rice batches.

Furthermore, an additional dataset was used for analyzing multiple grains present in a single image for grading purposes. As shown in Fig 4.2 and Fig 4.3, this dataset contains images organized into several grading categories including Premium, Grade1, Grade2, Grade3, Grade4, and Grade5, along with defect-related classes such as Broken rice, Chalky rice, Healthy rice and Discolored rice. Each class contains approximately 250–301 images, where multiple rice grains appear together in a single image rather than individual grain samples. This dataset helps represent real-world rice samples where many grains are present simultaneously, enabling analysis of grain distribution and quality variations within bulk rice images.

In addition, around 50 real-time images, as shown in Fig 4.4, were captured under varying lighting conditions (such as natural daylight, low light, and artificial illumination) to further enhance the robustness of the model. These images were augmented to 200 images to further training requirements. These images simulate practical deployment scenarios where lighting may not be controlled, thereby improving the system's ability to generalize across different environments. Together, these datasets provide a diverse collection of rice grain images covering both rice variety categories and grading-related classes, forming the complete dataset used in this study.

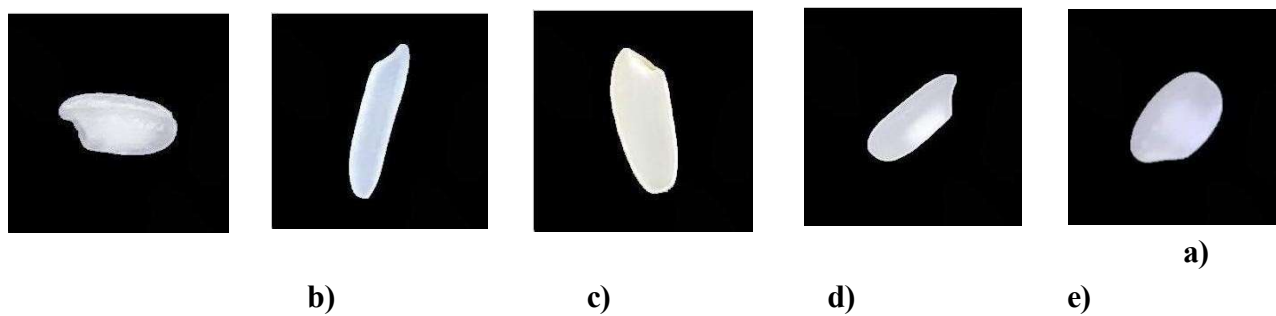


Fig 4.1 Rice varieties

a) Arborio, b) Basmati, c) Ipsala, d) Jasmine, e) Karacadag

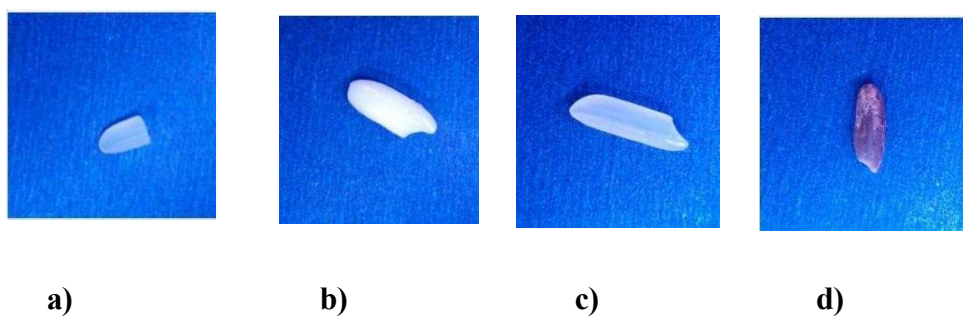


Fig 4.2 Defective and Healthy Rice images

a) Broken, b) Chalky, c) Healthy, d) Discoloured

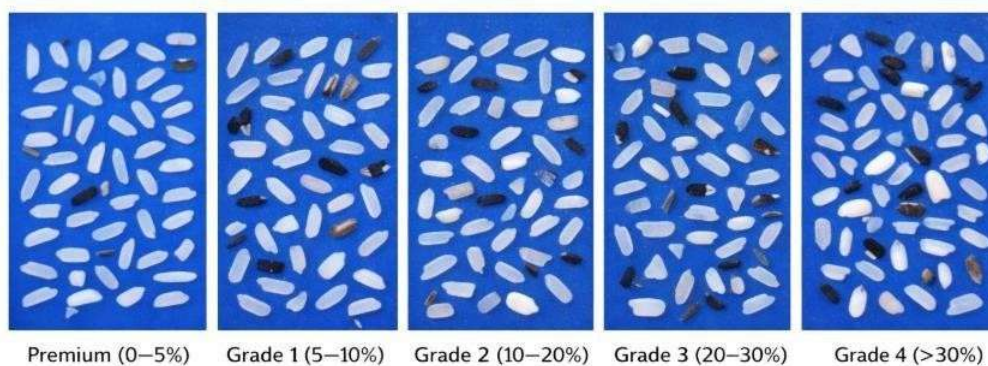


Fig 4.3 Graded Rice images



Fig 4.4 Real time rice grain images

4.3 DATA PREPROCESSING

Before training the classification model, several preprocessing steps were applied to prepare the rice grain images and ensure consistency in the dataset. For single grain images, preprocessing mainly involved dataset merging, downsampling, and data augmentation. Initially, multiple datasets were combined to create a unified dataset containing different rice grain varieties and defect categories. Since some classes contained a larger number of images compared to others, downsampling was applied to reduce the number of samples in dominant classes and maintain a balanced dataset distribution. In addition, data augmentation techniques were used to increase the diversity of the training data and improve the robustness of the model. Augmentation operations such as horizontal flipping (applied to 50% of the images) and small rotations within a range of $\pm 15^\circ$ (applied to around 30% of the images) were performed. A 50% probability for horizontal flipping was selected to ensure that approximately half of the images appear in a mirrored orientation, which helps the model learn grain features that are independent of left–right orientation while still preserving the original image distribution. Similarly, rotation was limited to $\pm 15^\circ$ and applied to around 30% of the images to simulate natural variations in grain orientation that occur during image capture, while avoiding excessive distortion that could alter the true shape characteristics of the grains. Applying rotation to a smaller percentage of images prevents the dataset from becoming overly transformed and helps maintain realistic grain structures. These preprocessing techniques help the model generalize better by exposing it to varied image patterns while preventing overfitting. After preprocessing, the images were resized and normalized so that they could be effectively used as input for the CNN classification model.

4.4 MODEL ARCHITECTURAL DESIGN

The proposed system consists of three main components for rice grain analysis. Contour-based segmentation is used to detect individual grains from bulk rice images using classical image processing techniques such as edge detection and contour extraction. In addition, a YOLOv8-based object detection approach is implemented to accurately detect and localize multiple grains in complex images using deep learning. After grain detection, the extracted single grain images are passed to a Convolutional Neural Network (CNN), which analyzes visual features such as shape and texture to classify the grains and support the rice grading process.

4.4.1 CONTOUR – BASED SEGMENTATION

In the contour-based approach, as shown in Fig. 4.5, a rice image with multiple rice grains is first provided as the input to the system. The process begins by converting the RGB image into a grayscale image, which simplifies the image representation and reduces computational complexity by removing color information that is not essential for boundary detection. Edge detection is then performed using the Canny edge detection algorithm to accurately highlight the boundaries of the rice grains, as this method is effective in detecting sharp intensity changes and preserving important edge details. After detecting the edges, morphological dilation is applied to close small gaps in the detected edges and connect fragmented boundaries, which helps form continuous grain outlines. This step improves segmentation by ensuring that the boundaries of each grain are clearly defined. Following this, contour extraction is performed to identify the outlines corresponding to potential rice grains in the image. This step improves segmentation by ensuring that the boundaries of each grain are clearly defined. Following this, contour extraction is performed to identify the outlines corresponding to potential rice grains in the image.

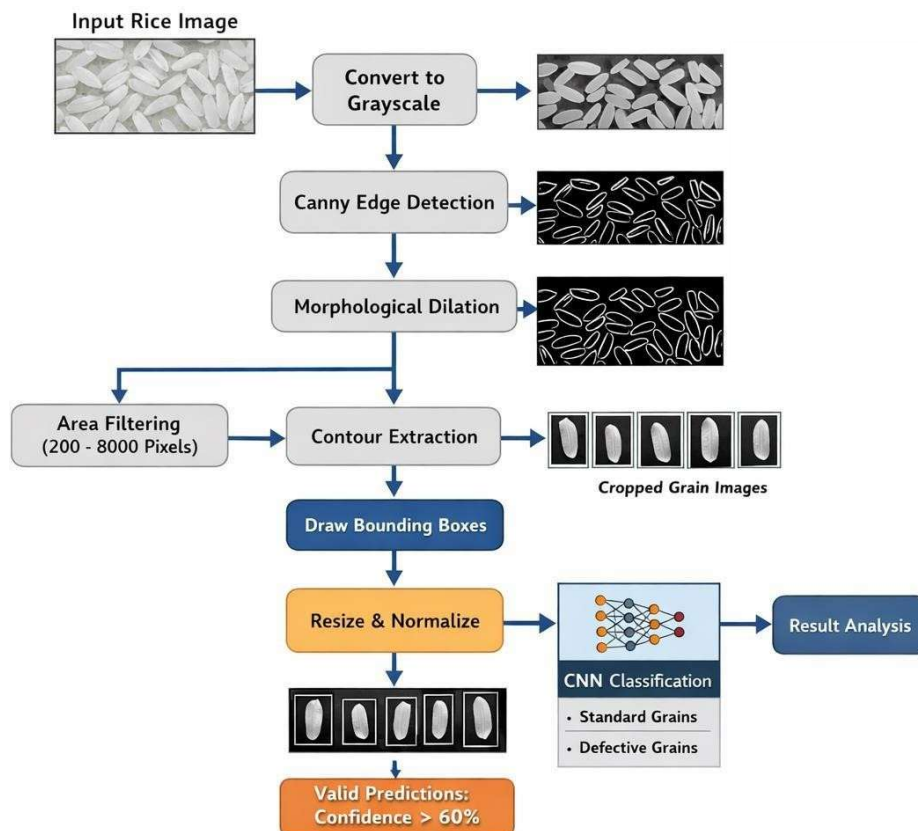


Fig 4.5 Rice grain quality grading design architecture – Contour based segmentation

To eliminate noise and irrelevant objects such as small particles or background artifacts, area filtering is applied with a threshold range of 200 to 8000 pixels. The lower threshold value of 200 pixels helps remove very small objects that are unlikely to be rice grains, while the upper threshold value of 8000 prevents large connected regions or overlapping grain clusters from being mistakenly detected as a single grain. Only contours within this range are considered valid rice grains. Once valid contours are identified, bounding boxes are generated around each detected grain, and these regions are cropped from the original image to obtain individual grain images. Each detected grain is then cropped, resized to match the input size required by the CNN model, and normalized, ensuring consistent input dimensions and pixel value scaling for reliable classification. The processed grain images are then passed to a Convolutional Neural Network (CNN) for classification. The CNN analyzes visual features such as grain shape, texture, and surface characteristics to classify each grain into categories such as standard or defective grains. After classification, the system counts the number of grains belonging to each category and calculates the percentage of defective grains present in the sample. Based on predefined grading criteria, the final rice quality grade is assigned to the sample. Additionally, during prediction, only classifications with a confidence score greater than 60% are considered valid results to ensure reliable grading and reduce incorrect predictions.

4.4.2 YOLOv8 – BASED SEGMENTATION

In the YOLOv8-based approach, as shown in Fig. 4.6, the process begins with a bulk rice image containing multiple grains. The image is provided to the YOLOv8 object detection model, which automatically detects and locates individual rice grains by generating bounding boxes around them. YOLOv8 was selected for this task because it is a real-time object detection algorithm capable of detecting multiple objects in a single image with high accuracy and speed, making it suitable for analyzing bulk rice samples where many grains appear simultaneously. In this approach, the training dataset was first prepared by manually annotating rice grain images with bounding boxes. Each grain present in the image was manually marked using an annotation tool, where rectangular bounding boxes were drawn around individual grains. These manually labeled images provide ground truth information that allows the YOLOv8 model to learn the spatial location, size, and shape of rice grains during the training process. Once the model is trained using these annotated images, it can automatically detect individual grains in new images.

This deep learning based detection method is highly effective in identifying grains even when they overlap or appear in complex backgrounds, which is often difficult for traditional image processing methods. Using the bounding box coordinates generated by YOLOv8, each grain region is cropped from the original image to obtain individual grain images.

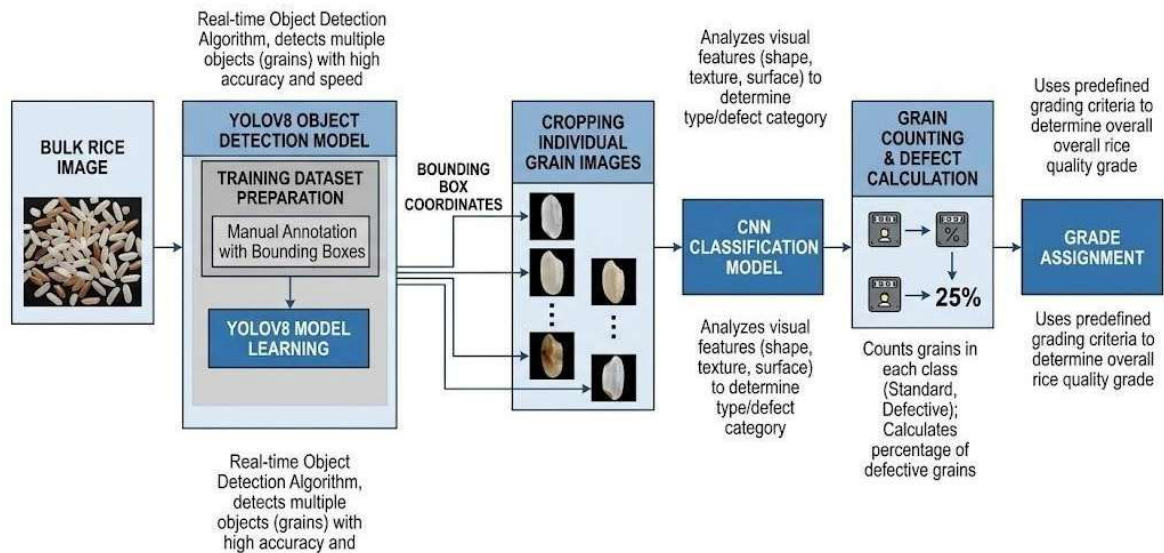


Fig 4.6 Rice grain quality grading design architecture – YOLOv8 based segmentation

These cropped grain images are then passed to a Convolutional Neural Network (CNN) classification model, which analyzes visual features such as grain shape, texture, and surface characteristics to determine the grain type or defect category. After classification, the system counts the number of grains belonging to each class, including standard and defective grains. The percentage of defective grains present in the sample is then calculated based on the total grain count. Finally, using predefined grading criteria, the system performs grade assignment, where the overall rice quality grade is determined. This approach enables accurate and efficient rice grading by combining the strong object detection capability of YOLOv8 with the classification ability of CNN, allowing grains to be analyzed individually from bulk images while maintaining high detection accuracy and processing efficiency.

4.4.3 CUSTOM CNN

After detecting individual rice grains using the contour-based and YOLOv8-based approaches, the extracted grain regions are passed to a Convolutional Neural Network (CNN) for classification. A lightweight CNN architecture with 10 layers was designed to classify the detected grains into different categories required for the grading process while keeping the computational cost low. The CNN model receives RGB rice grain images of size $224 \times 224 \times 3$ as input, which provides sufficient visual detail for capturing grain characteristics while maintaining efficient training and inference time. The network processes these images through multiple convolutional and pooling layers to extract important visual features. These layers allow the model to learn hierarchical representations of the grains such as edges, shapes, textures, and surface patterns, which are essential for distinguishing between different rice grain types and defect categories. The CNN architecture consists of 10 layers, including convolutional layers, activation layers, pooling layers, dropout layers, and fully connected layers. The convolutional layers are responsible for extracting spatial features from the input images, while the pooling layers reduce the spatial dimensions of the feature maps to lower computational complexity and retain the most important features. Dropout regularization with a rate of 0.4 is applied to reduce overfitting by randomly deactivating a portion of neurons during training, which improves the model's ability to generalize to unseen data. The final fully connected layer uses a Softmax activation function with output neurons, representing the rice grain classes used for classification, since Softmax converts the output values into probability scores suitable for multi-class prediction. During training, the network learns class-specific patterns by adjusting neuron weights through backpropagation. The model is trained using the Adam optimizer with a learning rate of 3×10^{-4} , as Adam provides faster convergence and stable training performance for deep learning models. Sparse Categorical Crossentropy is used as the loss function because it is well suited for multi-class classification problems with integer class labels.

4.4.4 LAYER-WISE CONFIGURATION

The Convolutional Neural Network (CNN) architecture is used for single grain classification in the proposed system which is shown in Fig 4.9. The CNN model analyzes cropped rice grain images obtained from the contour-based and YOLOv8-based grain detection stages.

Table 4.1 Model Architecture – Variety Classification

Block	Layer Details	Description / Purpose
Input Configuration	Input Shape: (224, 224,3)	RGB rice grain images are used as input to preserve spatial and color information.
Block 1 – Basic Feature Extraction	Conv2D (32 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Extracts low-level features such as edges and basic grain shape patterns while reducing spatial dimensions.
Block 2 – Intermediate Feature Learning	Conv2D (64 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Learns more complex features such as grain texture and structural variations between rice varieties.
Block 3 – Deep Feature Extraction	Conv2D (128 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Captures high-level abstract features such as grain surface characteristics and detailed shape patterns.
Flatten Layer	Flatten()	Converts the extracted feature maps into a one-dimensional vector for classification.
Classification Block	Dense (128 neurons, ReLU) Dropout (0.5) Dense (5 neurons, Softmax)	Fully connected layers perform classification into five rice varieties. Dropout reduces overfitting and Softmax produces probability scores for each class.

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_1 (Flatten)	(None, 86528)	0
dense_2 (Dense)	(None, 128)	11,075,712
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 5)	645

Fig 4.7 Network Architecture of the CNN Model Variety Classification

Table 4.2 Model Architecture – Defect Classification

Block	Layer Details	Description / Purpose
Input Configuration	Input Shape: (224, 224, 3)	RGB rice grain images are used as input to preserve spatial and color information for identifying defect characteristics.
Block 1 – Basic Feature Extraction	Conv2D (32 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Extracts low-level features such as grain edges, cracks, and basic structural patterns while reducing spatial dimensions.
Block 2 – Intermediate Feature Learning	Conv2D (64 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Learns intermediate features such as grain texture variations, surface irregularities, and minor damage patterns.
Block 3 – Deep Feature Extraction	Conv2D (128 filters, 3×3 kernel, ReLU) MaxPooling2D (2×2)	Captures high-level abstract features including broken grain structures, chalky regions, and discoloration patterns.
Flatten Layer	Flatten()	Converts the extracted feature maps into a one-dimensional feature vector for classification.
Classification Block	Dense (128 neurons, ReLU) Dropout (0.5) Dense (4 neurons, Softmax)	Fully connected layers perform classification into four defect classes. Dropout reduces overfitting and Softmax generates probability scores for each defect class (Broken, Chalky, Healthy, Discolored, Stone (Impurities)).

Layer (type)	Output Shape	Param #
conv2d_9 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_9 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_10 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_10 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_11 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_11 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_3 (Flatten)	(None, 86528)	0
dense_6 (Dense)	(None, 128)	11,075,712
dropout_3 (Dropout)	(None, 128)	0
dense_7 (Dense)	(None, 4)	516

Fig 4.8 Network Architecture of the CNN Model Defect Classification

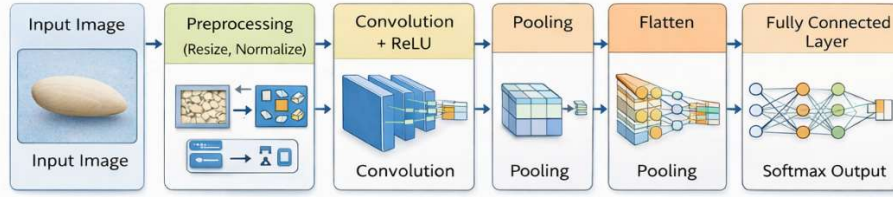


Fig 4.9 Model Architecture of the CNN Model (10 Layers)

These cropped images are provided as input to the CNN model, where the network extracts important visual features such as grain shape, surface texture, structural patterns, color variations, and damage characteristics. These features help the model distinguish between different rice varieties as well as defect categories such as broken, chalky, healthy, impurities (stones) and discolored grains. The CNN learns these discriminative features through multiple convolutional and pooling layers, enabling accurate identification of grain types and defects. The detailed architecture of the CNN models used for variety classification and defect classification is presented in Table 4.1 and Table 4.2, while the corresponding network structures are illustrated in Fig. 4.7 and Fig. 4.8 respectively.

4.5 GRADING CRITERIA

After detecting and classifying individual rice grains, the system determines the overall rice quality by calculating the percentage of defective grains present in the sample. Each detected grain is classified as either healthy or defective, and the total number of defective grains is counted. The defect percentage is then computed by dividing the number of defective grains by the total number of grains detected in the image. This value represents the proportion of grains that do not meet quality standards and serves as the basis for determining the final rice grade. The defect percentage is calculated using the formula shown below in equation 4.1.

$$Defect \% = \frac{Number\ of\ Defective\ Grains}{Total\ number\ of\ Grains} \times 100 \quad (4.1)$$

Based on the calculated defect percentage, the system assigns a rice quality grade using predefined grading thresholds, as shown in Table 4.3. Samples with a very low percentage of defective grains are classified as Premium, indicating superior rice quality with minimal defects.

Grade 1 represents high-quality with a low defects, while Grade 2, Grade 3 indicate moderate quality with increasing levels of broken, chalky or discolored grains. Samples containing a high percentage of defective grains are categorized into lower grades, indicating reduced rice quality. This system helps in objectively evaluating rice quality based on the proportion of defective grains, enabling consistent and automated quality assessment without manual inspection.

Table 4.3 Grading Criteria

Defect %	Grade	Description
0–5%	Premium	Very high quality rice.
5–10%	Grade 1	High quality rice.
10–20%	Grade 2	Moderate quality.
20–30%	Grade 3	Lower quality rice.
>30%	Grade 4	Poor quality rice.

4.6 MODEL TRAINING

The training stage focuses on enabling the Convolutional Neural Network (CNN) models to learn discriminative visual features from rice grain images for both variety classification and defect classification. During training, the models learn to identify important characteristics such as grain shape, texture, surface patterns, and color variations by adjusting the weights of the network through backpropagation. Separate CNN models are trained for variety classification and defect classification, although both follow a similar architecture design with modifications in the output layer based on the number of classes.

4.6.1 VARIETY CLASSIFICATION

For rice variety classification, the CNN model is trained to distinguish between five rice varieties: Arborio, Basmati, Ipsala, Jasmine and Karacadag. The network receives resized RGB rice grain images as input and processes them through multiple convolutional and pooling layers

to learn hierarchical spatial representations. The convolutional layers automatically extract spatial features such as edges, contours, textures, and intensity variations present in the rice grain images. The first convolution layer with 32 filters captures low-level visual patterns such as edges, grain boundaries, and basic shape structures. This layer enables the network to understand the fundamental structure, orientation, and outline of individual rice grains. The second convolution layer with 64 filters learns more complex features such as grain texture and internal structural variations between rice varieties. The third convolution layer with 128 filters captures higher-level features including subtle differences in grain shape and surface characteristics that are important for distinguishing similar rice types. After each convolution layer, MaxPooling layers are applied to reduce the spatial dimensions of the feature maps. This helps decrease computational complexity while retaining the most important visual information. Pooling also improves the robustness of the model by making the learned features less sensitive to small positional variations in the input images. The extracted feature maps are then passed through a Flatten layer, which converts the two-dimensional feature maps into a one-dimensional feature vector that can be processed by fully connected layers. The classification stage consists of a Dense layer with 128 neurons, which performs high-level feature learning by combining the extracted features to identify the most relevant patterns associated with each rice variety. A Dropout layer is included to reduce overfitting by randomly deactivating neurons during training, allowing the model to generalize better to unseen data. Finally, a Softmax output layer with five neurons is used to produce probability scores corresponding to the five rice varieties. The Softmax function ensures that the output values represent normalized probabilities for multi-class classification.

4.6.2 CONTOUR AND YOLOv8 TRAINING

In the contour-based approach, grain detection is performed using classical image processing techniques rather than a deep learning model. Therefore, no separate training process is required for this method. The segmentation pipeline consists of several preprocessing steps including grayscale conversion, edge detection, morphological operations, and contour extraction. The Canny edge detection algorithm is used to highlight grain boundaries, while morphological dilation is applied to close small gaps in the detected edges. After preprocessing, contours representing individual grains are extracted and filtered based on their area to remove noise and non-grain objects. To ensure reliable grain detection, area filtering thresholds were applied between 200 and 8000 pixels, allowing the system to ignore very small noise particles and large merged grain

regions. Bounding boxes are then generated around the detected contours, and individual grains are cropped from the original image for further classification using the CNN model. Since this approach does not involve learning from data, performance evaluation is mainly based on the accuracy of grain segmentation and the quality of the cropped grain images used for defect classification. In the YOLO-based approach, grain detection is performed using the YOLOv8 deep learning object detection model. Unlike the contour method, this approach requires a training process in which rice grain images are manually annotated with bounding boxes. During dataset preparation, bounding boxes were drawn around each individual grain using an annotation tool, and these labeled images were used to train the YOLOv8 model to learn the spatial location and shape of rice grains in bulk images. The trained model is capable of detecting multiple grains in a single image, even when grains overlap or appear in complex backgrounds. The YOLOv8 model was trained using standard object detection evaluation metrics including Precision, Recall, and Mean Average Precision (mAP) as shown in equation 4.2, 4.3, 4.4 and 4.5.

$$Precision = \frac{TP}{TP + FP} \quad (4.2)$$

$$Recall = \frac{TP}{TP + FN} \quad (4.3)$$

$$Avg\ Precision = \int_0^1 Precision(Recall)d(Recall) \quad (4.4)$$

$$Mean\ Average\ Precision = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4.5)$$

These metrics are commonly used in object detection tasks because they measure both the accuracy of detected bounding boxes and the ability of the model to correctly identify objects. The bounding box coordinates generated by the YOLO model are then used to crop individual grain images, which are passed to the CNN classifier for defect and variety classification.

4.6.3 DEFECT CLASSIFICATION

For defect classification, a similar CNN architecture is used to classify rice grains into defect categories such as Broken, Chalky, Healthy, and Discolored grains. The model receives cropped rice grain images obtained from the grain detection stage, where grains are first detected

using either the contour-based segmentation approach or the YOLOv8-based object detection model. These extracted grain images are then provided as input to the CNN model for defect classification. The network processes the images through multiple convolutional layers designed to capture defect-related visual patterns present in the grains. The initial convolution layers extract low-level features such as grain edges and crack patterns, while deeper convolution layers learn more complex characteristics including discoloration regions, chalky surfaces, and irregular grain structures that indicate defects.

The use of multiple convolution layers allows the network to learn hierarchical features, progressing from simple edge detection to complex texture recognition. MaxPooling layers are applied after convolution operations to reduce the spatial dimensions of feature maps while preserving the most important visual information. The Flatten layer converts the extracted feature maps into a one-dimensional feature vector, which is then processed by a fully connected Dense layer for classification. To improve model generalization and prevent overfitting, a Dropout layer is applied before the final classification layer. The output layer uses a Softmax activation function with four neurons, corresponding to the four defect classes. This architecture enables the model to learn discriminative visual patterns associated with rice grain defects and accurately classify each grain, supporting the automated rice quality grading framework.

4.6.4 PERFORMANCE TRACKING

During training, the performance of the CNN model for rice variety classification was monitored by recording the training accuracy and loss values for each epoch to evaluate the learning progress of the network. The model was trained to classify rice grains into five rice varieties: Arborio, Basmati, Ipsala, Jasmine, and Karacadag. Throughout the training process, the accuracy gradually increased while the loss decreased, indicating that the model effectively learned the distinguishing visual characteristics of different rice varieties such as grain shape, texture, and structural patterns.

To further evaluate the model performance, several classification metrics including accuracy, precision, recall, and F1-score were used. Confusion matrices and classification reports were generated to analyze the class-wise prediction results and ensure balanced classification performance across all rice variety classes. For grain detection, two approaches were evaluated. The contour-based approach does not involve model training and its effectiveness was assessed

based on the quality of grain segmentation and the accuracy of extracted grain regions. In contrast, the YOLOv8 detection model was trained using manually annotated bounding box images of rice grains. The performance of the YOLO model was evaluated using standard object detection metrics including Precision, Recall, and Mean Average Precision (mAP). The trained model achieved high detection performance with Precision = 0.998, Recall = 1.000, and mAP@50 = 0.995, indicating highly accurate localization of individual grains in bulk rice images. For defect classification, a CNN model was trained to classify rice grains into four defect categories: Broken, Chalky, Healthy, and Discolored. After preprocessing, the dataset was split into training and validation sets. All images were resized to 224×224 pixels and normalized before training. The model was trained for 30 epochs using the Adam optimizer with a learning rate of 0.0001, while sparse categorical cross-entropy was used as the loss function for multi-class classification. The model achieved a training accuracy of approximately 93–94% and a validation accuracy of about 94.3%, indicating stable learning performance. Further evaluation using precision, recall, and F1-score showed values ranging between 0.92 and 1.00 across the different defect classes, confirming balanced and reliable classification performance. These metrics demonstrate that the CNN model can effectively identify rice grain defects with high accuracy and minimal misclassification. Overall, the results indicate that the proposed detection and classification framework provides robust and reliable performance for automated rice grain grading and analysis.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 MODEL TRAINING AND EVALUATION

5.1.1 SINGLE GRAIN PREDICTION

5.1.1.1 RICE TYPE PREDICTION RESULTS

The proposed CNN model was trained on rice grain images of 5 different rice types using an 80:10:10 split for training, validation, and testing. With 10 layers, it achieved a 97% validation accuracy and a 0.0184 validation loss. A comparative analysis with 8 layer which achieved 96.48% train accuracy and 88.33% validation accuracy is shown in Fig. 5.1 (a–d), highlighting the accuracy and loss variations.

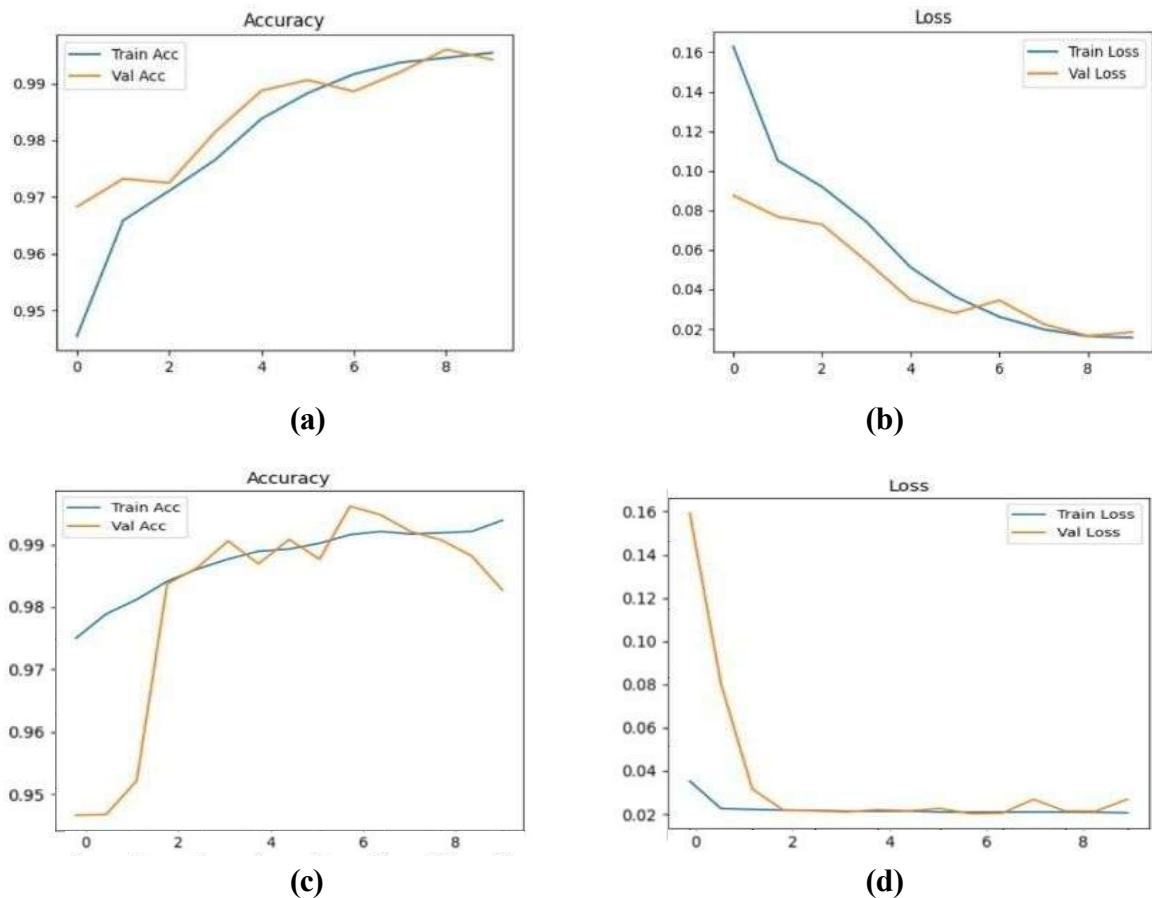


Fig 5.1 a) Accuracy plot of CNN (10 layers), b) Loss plot of CNN (10 layers),
c) Accuracy plot of CNN (8 layers), d) Loss plot of CNN (8 layers).

To further evaluate the performance, the same preprocessed rice grain dataset was used to train two pre-trained models, ResNet50 and EfficientNetB0. The ResNet50 model achieved a validation accuracy of 86.59%, while EfficientNetB0 obtained 88.77%. The relatively lower performance of these models is mainly because rice grains are small objects with very subtle differences in texture, shape, and color, making feature extraction more challenging. Additionally, these pre-trained models are originally trained on large natural image datasets like ImageNet, which contain general objects rather than fine-grained rice grain patterns. Due to this domain difference and the limited size of rice-specific datasets, the models may not adapt effectively, resulting in lower accuracy compared to a custom CNN designed specifically for rice grain features.

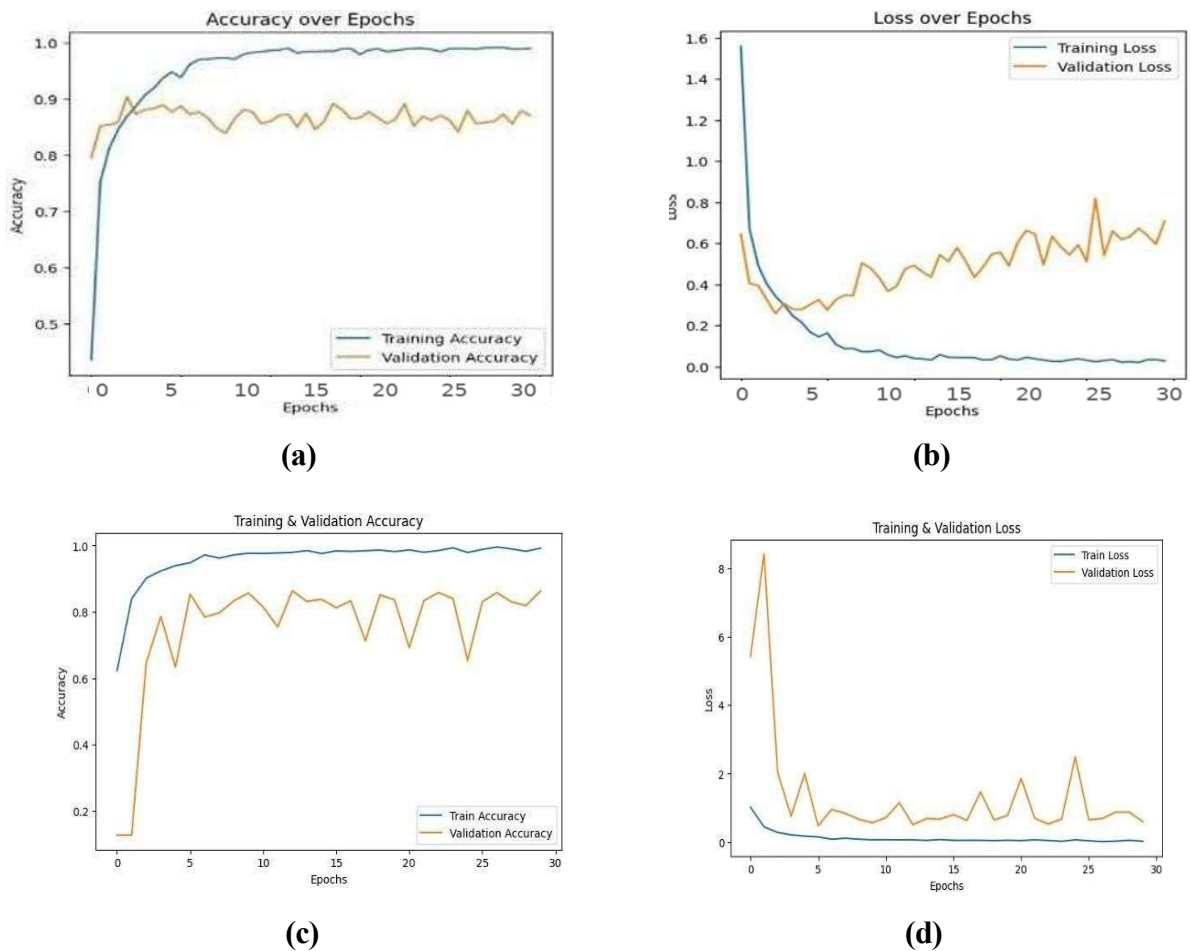


Fig 5.2 a) Accuracy plot of ResNet50, b) Loss plot of ResNet50, c) Accuracy plot of EfficientNetB0, d) Loss plot of EfficientNetB0.

The confusion matrices for the models CNN with 10 layers, CNN with 8 layers, ResNet50, and EfficientNetB0 are shown in Fig. 5.3 (a–d). These matrices illustrate how each model performs in classifying the data. By comparing the correctly and incorrectly predicted values, it becomes possible to clearly identify which model achieves higher accuracy and where each one tends to make errors.

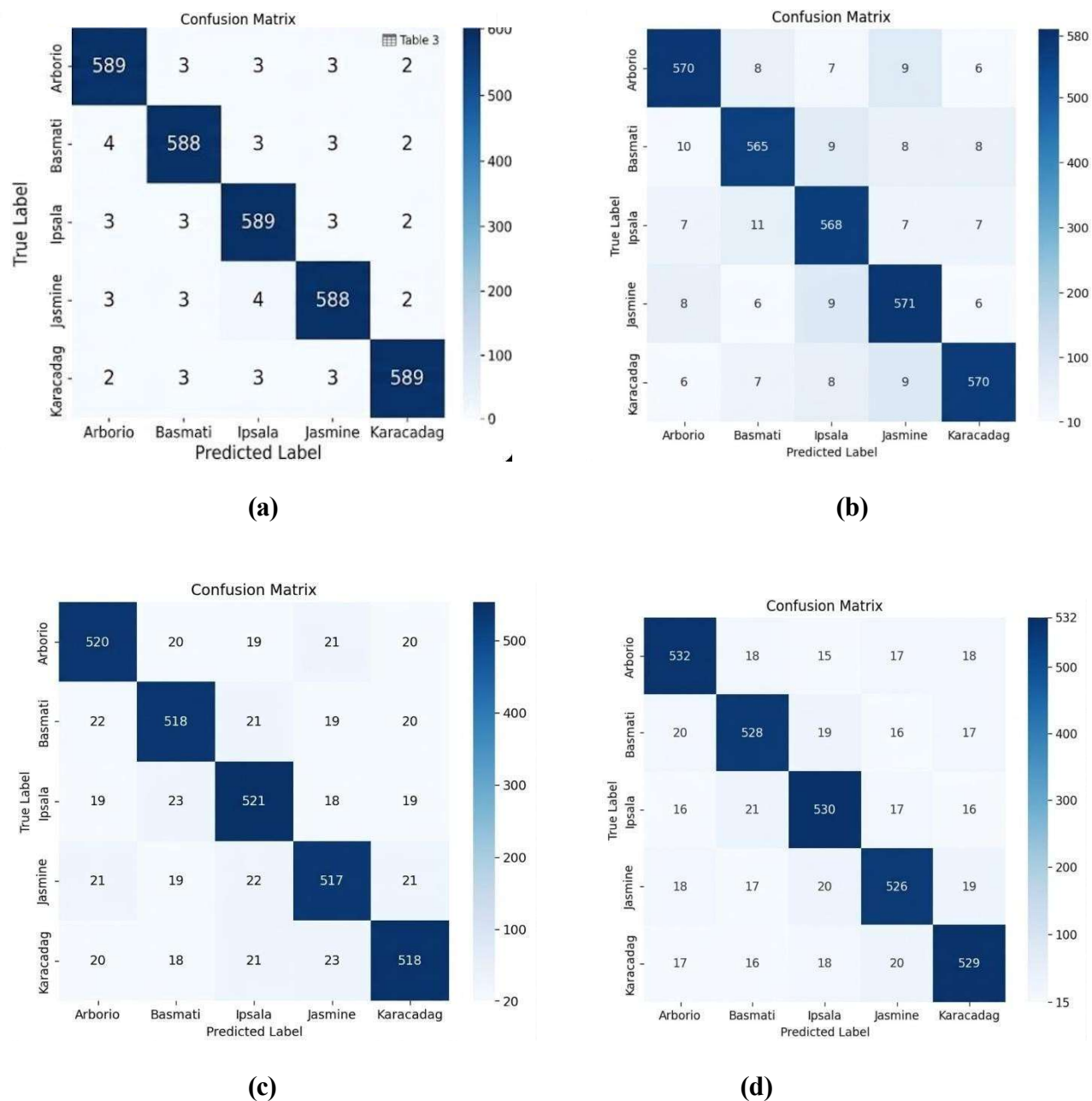


Fig 5.3 Confusion matrices of a) CNN (10 layers), b) CNN (8 layers), c) ResNet50, d) EfficientNetB0.

5.1.1.2 COMPARATIVE ANALYSIS

The comparative analysis of the various models - CNN (10 layers), CNN (8 layers), Resnet50, and EfficientNetB0 is presented below in Table 5.1. This comparison focuses on the training, validation, and testing accuracies of each model to evaluate their overall performance and generalization ability. By analyzing these metrics side by side, the learning behavior of each model and its performance on unseen data can be evaluated.

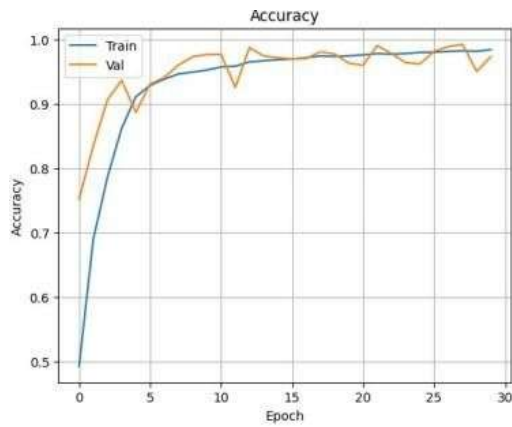
Table 5.1 Comparative analysis for Rice type Prediction

Model Name	Train (%)	Validation (%)	Test (%)
CNN (10 Layers)	98.37	97.87	96.8
CNN (8 Layers)	96.48	88.33	89.4
ResNet50	93.85	86.59	88.16
EfficientNetB0	99.21	88.77	85.75

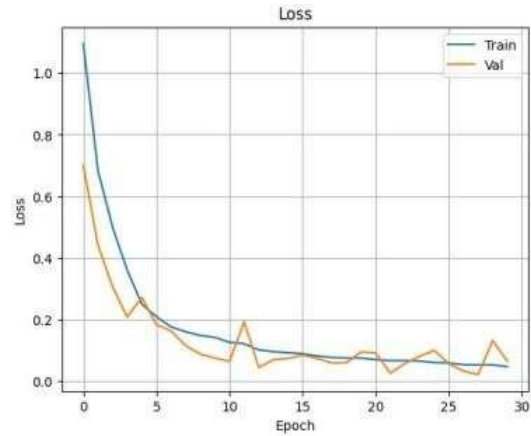
Models with high training accuracy but lower validation or testing accuracy indicate overfitting, while those maintaining consistent accuracy across all phases demonstrate better generalization. Through this comparative study, the differences in learning behavior and efficiency between simple CNN architectures and advanced models like ResNet50 and EfficientNetB0 become more apparent, helping to identify the most robust and reliable model for the given task.

5.1.1.3 RICE DEFECT PREDICTION RESULTS

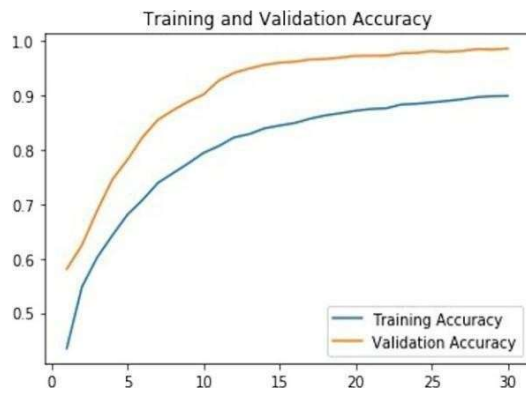
The proposed CNN model was trained on preprocessed images using an 80:10:10 data split. With 10 layers, the model achieved a validation accuracy of 98.48% and a validation loss of 0.0177. Comparative analysis with CNN model of 10 layers, 8 layers and pre-trained model EfficientNetB0 was also conducted, with accuracy and loss plot and confusion matrix shown in Fig 5.4 (a-f) and 5.5 (a-c).



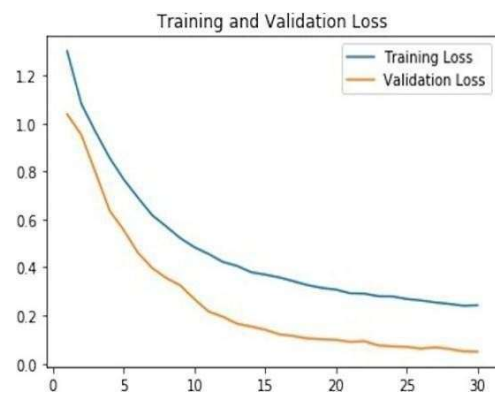
(a)



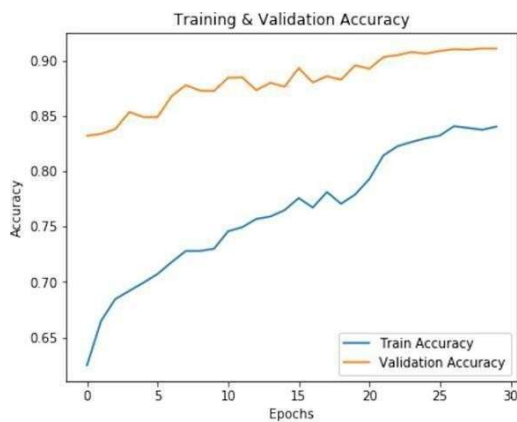
(b)



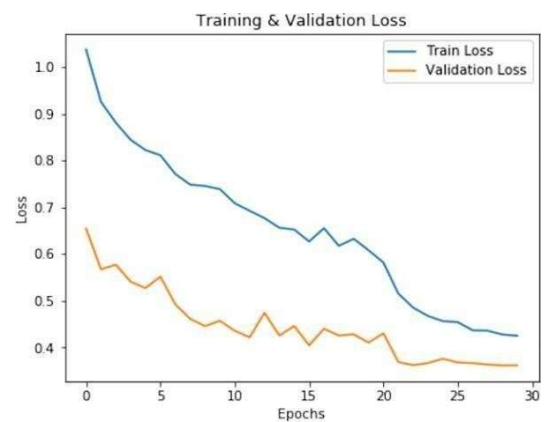
(c)



(d)

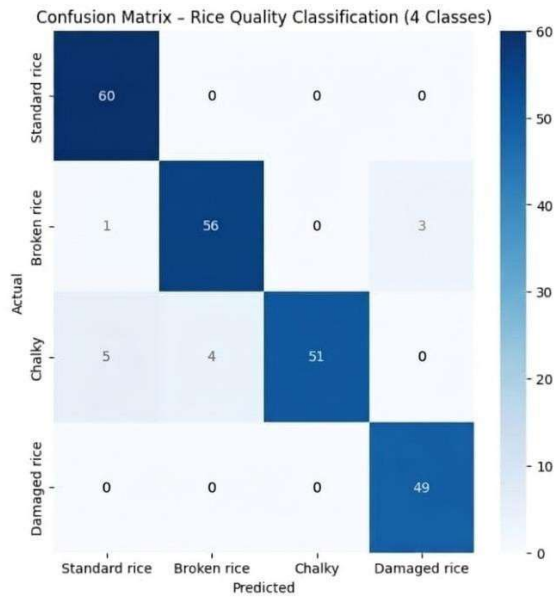


(e)

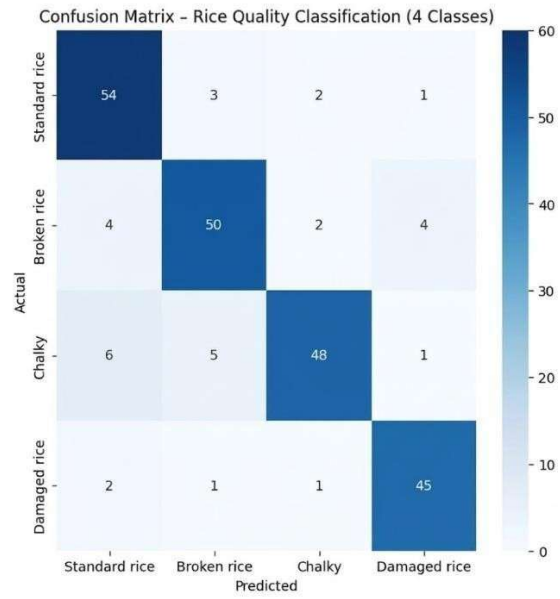


(f)

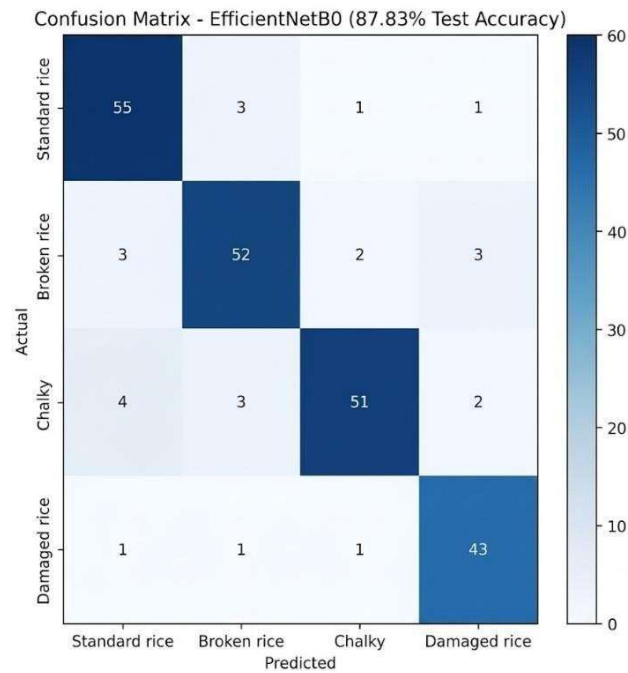
Fig 5.4 a) Accuracy plot of CNN (10 layers), b) Loss plot of CNN (10 layers), c) Accuracy plot of CNN (8 layers), d) Loss plot of CNN (8 layers), e) Accuracy plot of EfficientNetB0, f) Loss plot of EfficientNetB0.



(a)



(b)



(c)

Fig 5.5 Confusion matrices of a) CNN (10 layers),
b) CNN (8 layers), c) EfficientNetB0.

5.1.1.4 COMPARATIVE ANALYSIS

The comparative analysis of three CNN architectures, 10-layer CNN, 8-layer CNN, and EfficientNetB0, is presented in Table 5.2. This evaluation focuses on the training, validation, and testing accuracies of each model to assess overall performance and generalization capability. By comparing these metrics, the analysis highlights how model architecture and network depth influence learning capacity, and identifies the configuration that provides the best balance between model complexity, accuracy, and generalization performance.

Table 5.2 Comparative analysis for Rice Defect Analysis

Model Name	Train (%)	Validation (%)	Test (%)
CNN (10 Layers)	98.48	97.37	97.78
CNN (8 Layers)	96.21	88.33	85.83
EfficientNetB0	95.84	88.77	87.43

The comparative study of the three architectures shows clear differences in classification performance. The 10-layer CNN, which serves as the proposed model, achieved the highest overall accuracy, demonstrating strong capability in learning discriminative features from rice grain images while maintaining good generalization on unseen data. The 8-layer CNN also performed reasonably well but showed slightly lower accuracy due to its relatively shallow architecture, which limits its ability to capture more complex feature representations. In contrast, EfficientNetB0 achieved comparatively lower performance under the same training conditions. This may be attributed to the fact that pre-trained architectures are originally optimized for large natural image datasets, making them less effective in capturing the subtle texture and shape variations present in small rice grain images. Overall, the results indicate that the custom 10-layer CNN provides the best balance between accuracy of 97.78%, computational efficiency, and feature learning capability for rice grain classification.

5.1.2 MULTIPLE GRAIN PREDICTION

5.1.2.1 BASED ON CONTOUR SEGMENTATION

Fig 5.6 (a – b) illustrates sample outputs of the proposed multi-grain analysis pipeline based on contour segmentation. The system first detects individual rice grains from bulk images using contour extraction and generates bounding boxes around each detected grain. Each segmented grain is then passed to the trained CNN model for classification into standard, broken, chalky, impurities and damaged rice categories. Based on the class-wise counts, the system computes the defect percentage and assigns the final rice quality grade according to the predefined grading criteria.



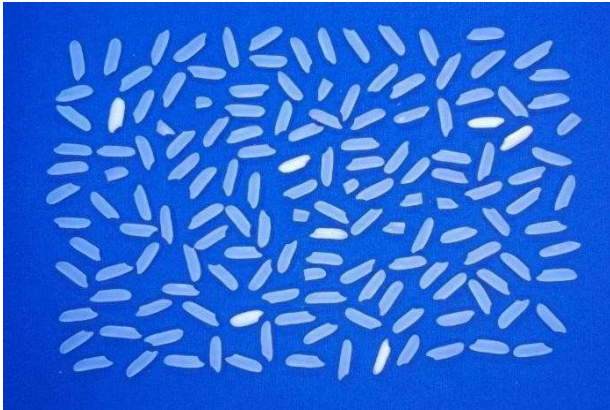
Fig 5.6 Predicted Outputs based on Contour Segmentation

a) Grade 1, b) Premium Grade

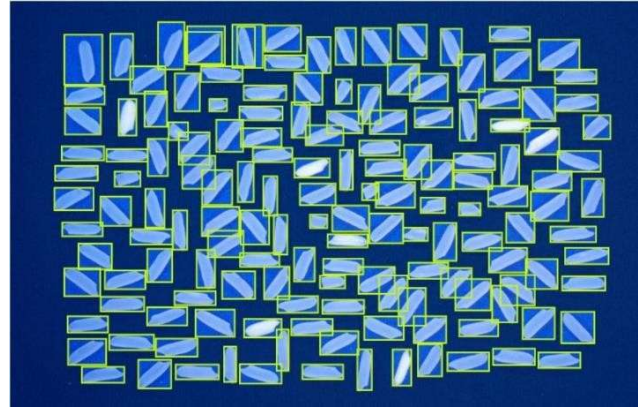
In the Fig 5.6 a), 137 grains were detected with 13 chalky grains identified as defective resulting in a 9.49% defect percentage, which corresponds to Grade 1. In the Fig 5.6 b), 141 grains were detected with only 1 defective grain producing a 0.71% defect percentage, leading to a Premium quality grade. These results demonstrate that the contour segmentation approach can effectively detect multiple grains in bulk images, automatically estimate the overall rice quality.

5.1.2.2 YOLO-BASED MULTI-GRAIN DETECTION

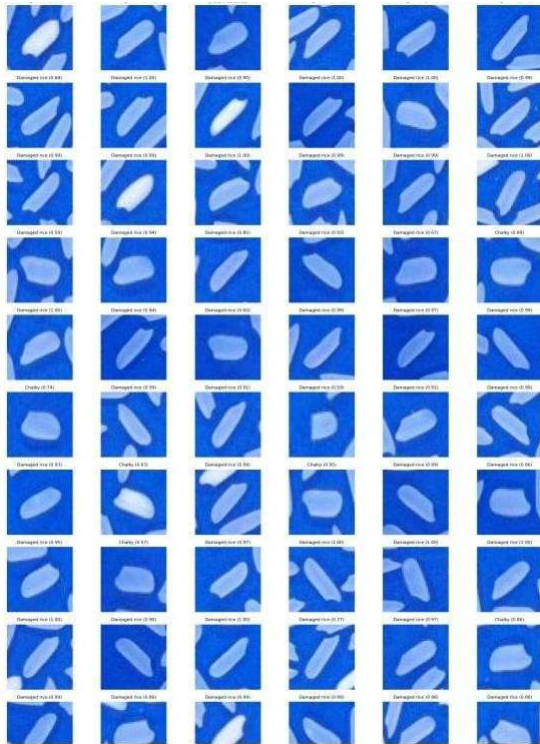
Fig 5.7 (a - d) illustrates the performance of the proposed YOLOv8-based multi-grain detection and grading pipeline. Fig 5.7 a) shows the original bulk rice image before detection, where numerous grains are present without any annotations. After applying the YOLOv8 detection model, Fig 5.7 b) displays the detection results in which bounding boxes are generated around each individual grain. The model demonstrates strong detection capability with Precision (P) = 0.998 and Recall (R) = 1.000, indicating that almost all predicted detections are correct and that the model successfully detects nearly every grain present in the image. The detection accuracy is further validated by the mAP@50 score of 0.995, which shows excellent detection performance at an IoU threshold of 0.5, while the mAP@50–95 value of 0.903 indicates consistent performance across multiple IoU thresholds. These results confirm that the YOLOv8 model can detect small rice grains even in dense and cluttered images. Following detection, the bounding boxes generated by YOLOv8 are used to extract individual grain regions from the original image, as shown in Fig 5.7 c). Each cropped grain image is resized and normalized before being passed to the trained CNN classifier for defect classification. The CNN model analyzes visual features such as grain texture, color variation, and shape patterns to classify each grain into categories including standard, broken, chalky, and damaged rice. Based on the classification results, the system performs class-wise grain counting and calculates the percentage of defective grains present in the image. Using the predefined grading thresholds, the system then determines the overall rice quality grade automatically. As illustrated in Fig 5.7 d), the final prediction output displays the detected grains with their corresponding labels along with the computed quality grade. In this example, several defective grains are identified, resulting in a final quality classification of Grade 4. The integration of YOLOv8 for accurate grain localization and CNN for defect classification enables efficient grain-level analysis and improves the reliability of automated rice quality grading. This approach demonstrates the capability of the proposed system to perform real-time bulk rice inspection with consistent and objective quality evaluation.



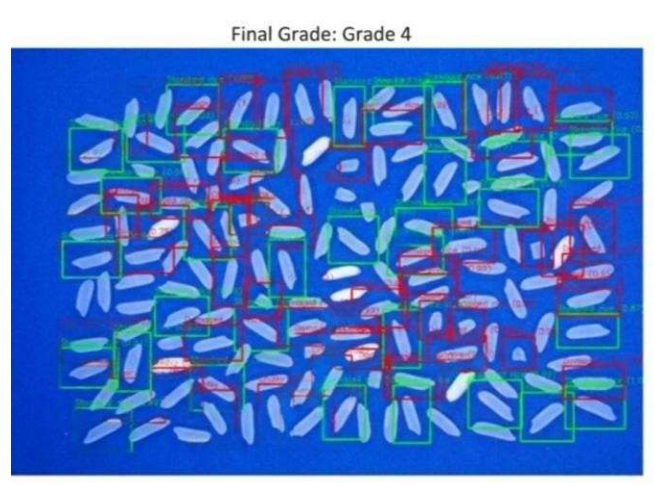
(a)



(b)



(c)



(d)

Fig 5.7 Rice Grain a) Before Bounding Boxes, b) After Bounding Boxes, c) Segmented grains d) Result after Prediction

5.1.2.3 COMPARATIVE ANALYSIS

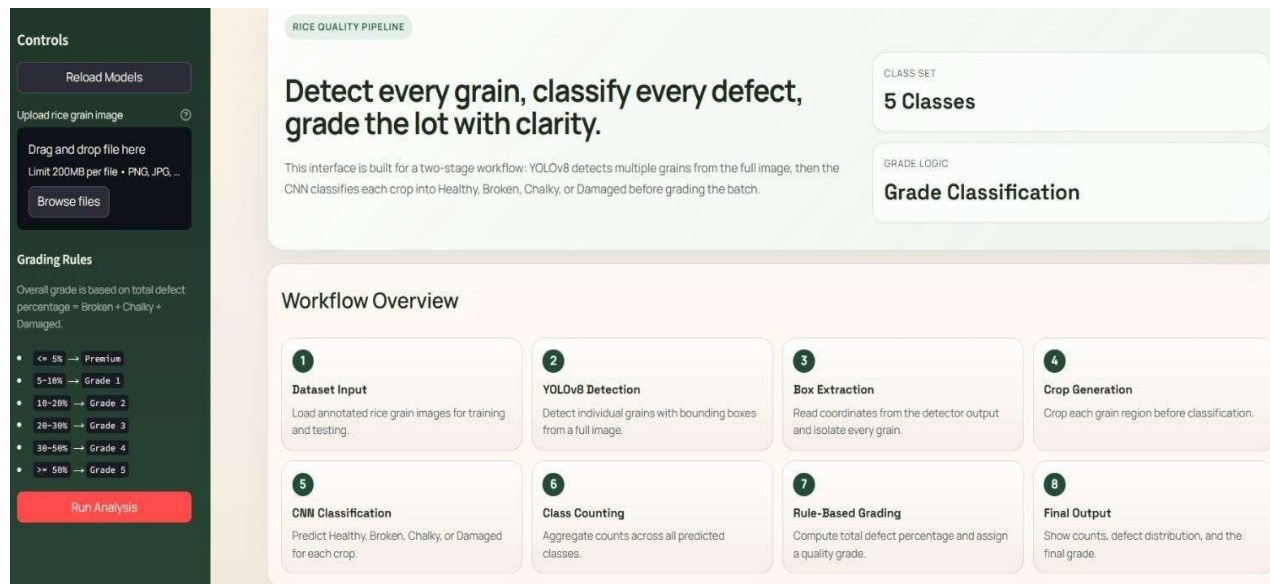
Contour segmentation and YOLOv8 represent two different approaches for detecting rice grains in bulk images. Contour segmentation is based on traditional image processing techniques such as grayscale conversion, edge detection, morphological operations, and contour extraction to identify grain boundaries. This method is lightweight and requires no training data, making it easy to implement. However, its performance strongly depends on consistent lighting conditions, clear grain boundaries, and minimal overlap between grains. When grains are densely packed or overlapping, the contour method may detect multiple grains as a single contour, which can affect grain counting and segmentation accuracy. In contrast, YOLOv8 is a deep learning-based object detection model that learns visual features directly from annotated datasets. Instead of relying on handcrafted image processing rules, YOLOv8 automatically identifies grain patterns, shapes, and textures during training. This allows it to detect grains even in complex scenes where grains are closely clustered or partially overlapping. Additionally, YOLOv8 is more robust to variations in lighting, orientation, and background conditions. Due to its ability to learn complex features and adapt to different image conditions, YOLOv8 provides a more reliable and scalable solution for multi-grain detection and automated rice quality analysis compared to traditional contour-based segmentation method.

5.2 MODEL DEPLOYMENT

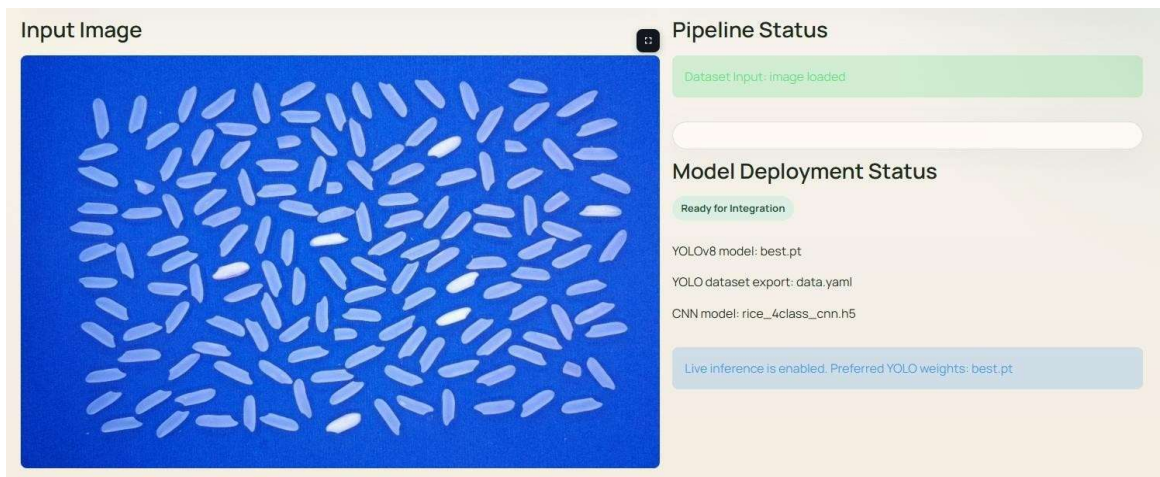
The project includes a Streamlit-based web application for automated rice grain quality analysis using a hybrid YOLOv8 and CNN approach. Initially, a trained YOLOv8 object detection model is used to detect and localize individual rice grains in the uploaded image. Each detected grain is then segmented using the bounding box regions and passed to a CNN classifier for fine-grained defect classification. The CNN assigns labels such as Healthy, Broken, Chalky, Discoloured, and Stones to each individual grain. Based on the CNN classification results, the system computes the total number of grains, along with the number of defective grains in each category. It then calculates class-wise percentages and the overall defect percentage. Using these metrics, a rule-based grading mechanism is applied to classify rice quality into categories ranging from Premium to Grade 5. The application also presents detailed outputs, including per-class defect counts, total grain count, annotated images, providing interpretable quality assessment. This model deployment and predicted outputs are shown in Fig 5.8 (a), (b), (c).

5.3 OUTPUT

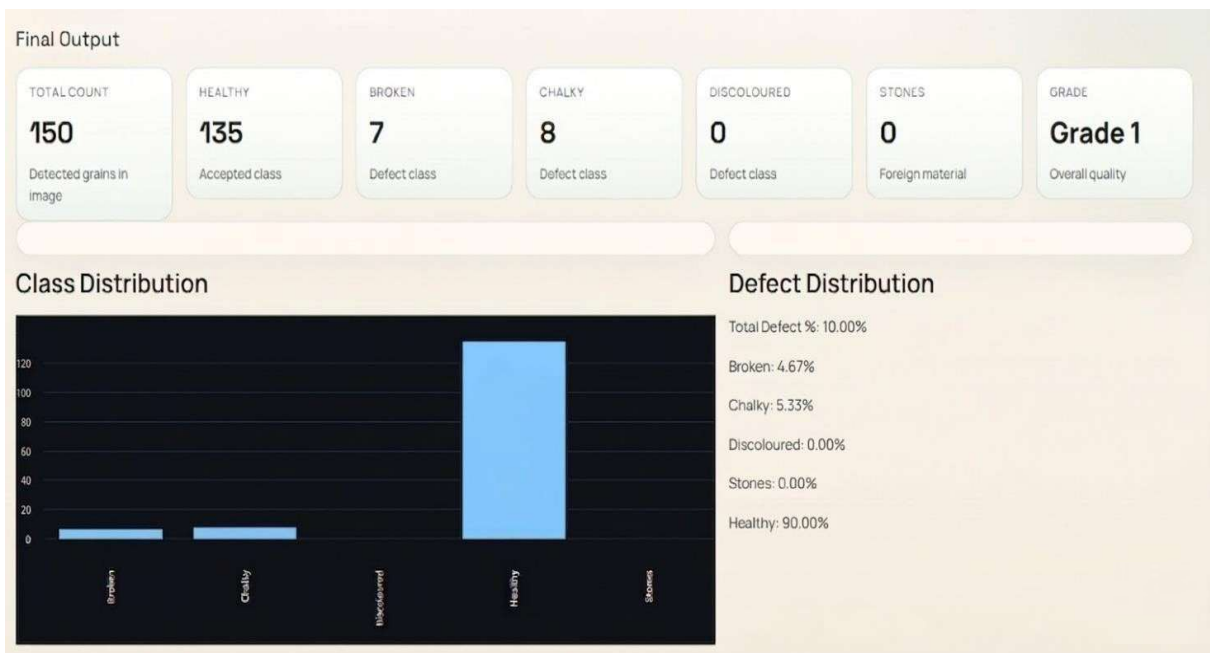
The real-time outputs demonstrate the effective functioning of the proposed rice grain quality analysis system across four different classes of rice samples. The trained YOLOv8 and CNN models were deployed through a Streamlit interface, allowing users to input rice grain images captured using a mobile camera. The system detects individual grains using YOLOv8, segments them, and classifies each grain using the CNN model into categories such as Healthy, Broken, Chalky, Discoloured, and Stones. It was tested on four distinct sample image sets representing different quality conditions, producing consistent predictions for grain classification, total grain count, and defect distribution. For images with good lighting and clarity, the system achieved accurate and stable results across all classes, while minor variations in performance were observed in some cases due to changes in image quality and lighting conditions. Overall, the system provides reliable outputs, including per-class defect counts, total number of grains, and final quality grading, demonstrating its effectiveness for real-time rice quality assessment. The Fig 5.9 (a), (b) represent real-time outputs obtained from mobile camera-based image inputs.



(a)

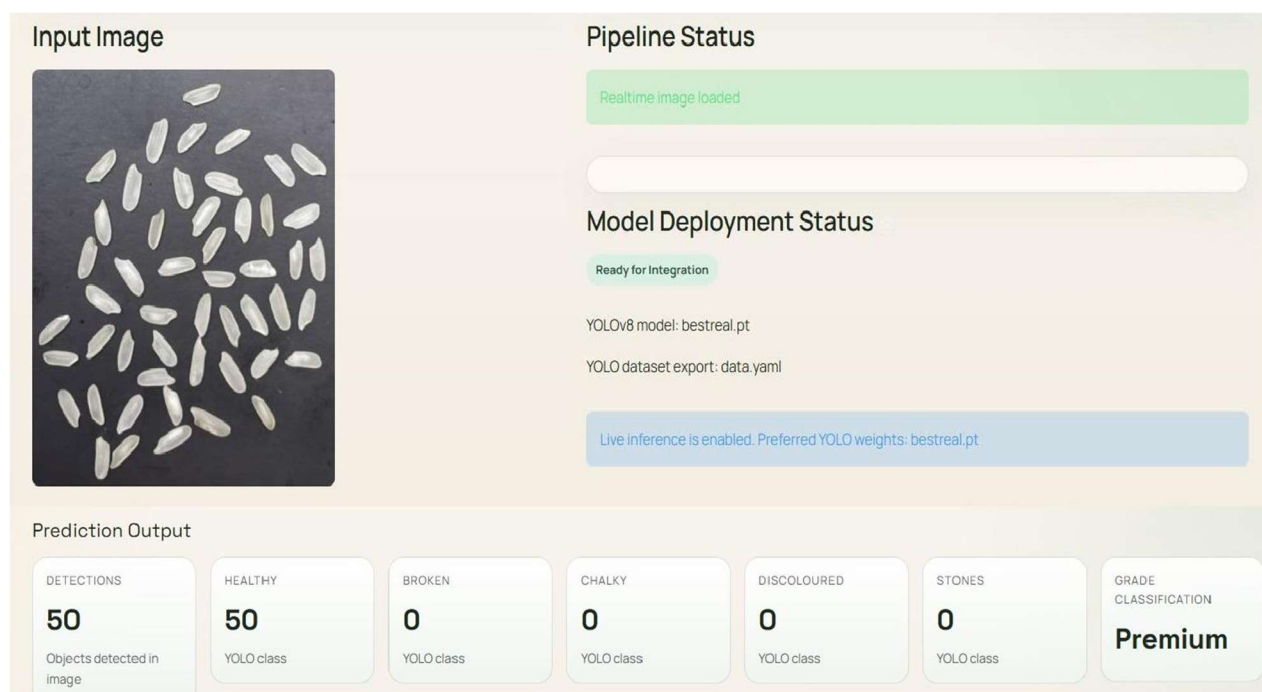


(b)

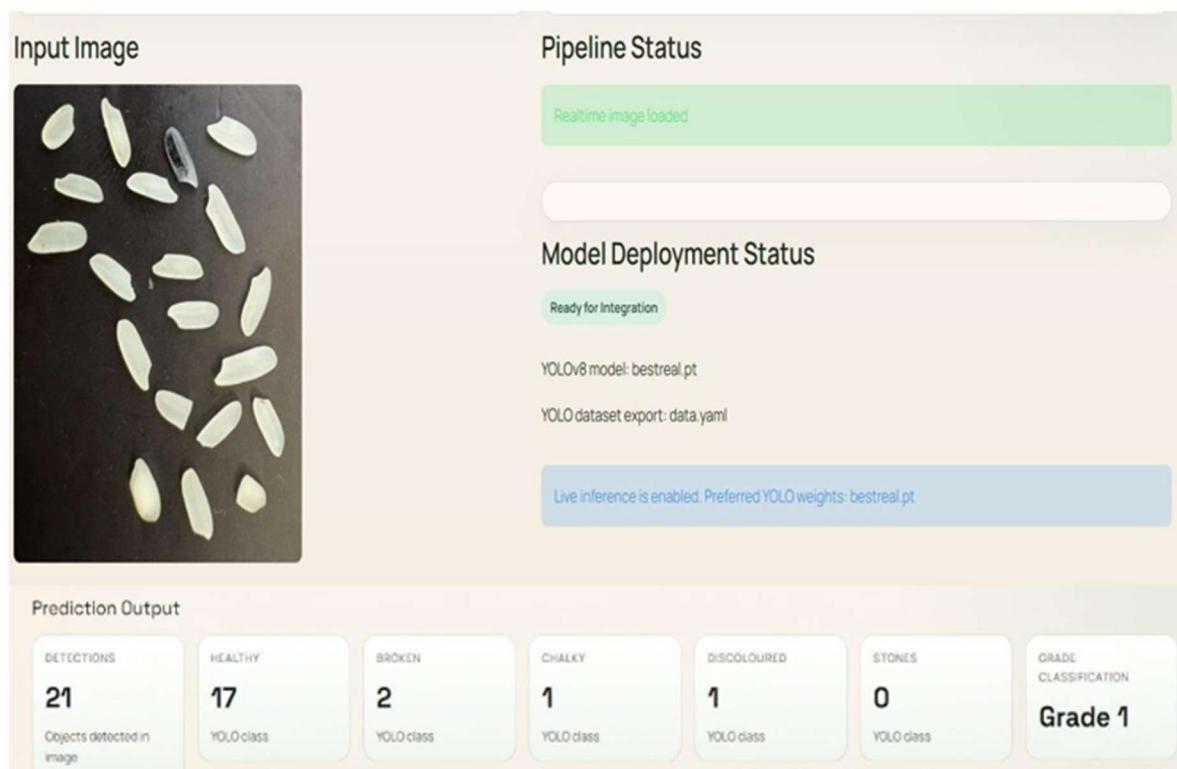


(c)

Fig 5.8 Model a) Deployment Interface,
 b) Input, c) Output.



(a)



(b)

Fig 5.9 Predicted Outputs for a) Premium, b) Grade 1.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

This project presents an automated rice quality analysis system using deep learning and computer vision techniques to detect grain defects, classify rice varieties, and determine overall rice quality from bulk images. A custom 10-layer Convolutional Neural Network (CNN) was developed and trained for both rice variety classification and defect classification using rice grain images. The model achieved a training accuracy of 98.48%, validation accuracy of 97.37%, and testing accuracy of 97.78% for classifying rice grains into categories such as healthy rice, broken rice, chalky grains, discolored or damaged rice, along with impurities such as stones. In addition, the model effectively identifies different rice varieties, demonstrating its capability to learn fine-grained visual features. These results indicate that the CNN architecture effectively learns discriminative features related to grain texture, shape, and color variations. The performance of the proposed model was also compared with other architectures, and it demonstrated a strong balance between accuracy and generalization. For multi-grain analysis, two grain detection approaches were evaluated. The first approach used contour-based segmentation, which detects grain boundaries through image processing techniques. Experimental results showed that this method can detect and segment multiple grains and estimate the defect percentage to determine rice quality grades. For example, in one test sample, 137 grains were detected with 13 defective grains, resulting in a defect percentage of 9.49% (Grade 1), while another sample with 141 grains contained only one defective grain, resulting in a Premium grade classification. To improve detection performance in dense grain arrangements, a YOLOv8-based object detection model was implemented. The model achieved excellent detection performance with Precision of 0.998, Recall of 1.000, mAP@50 of 0.995, and mAP@50–95 of 0.903, demonstrating its ability to accurately detect small rice grains even in cluttered scenes. The detected grains were then classified using the trained CNN model to compute the percentage of defective grains and determine the final rice quality grade automatically, along with identifying the rice variety. In addition, the system was tested on real-time images, including samples containing impurities such as stones, using a

Streamlit-based web application. The model successfully detected grains, classified varieties and defects, and identified impurities under varying lighting conditions and real-world scenarios, demonstrating its practical applicability beyond controlled datasets. Overall, the experimental results demonstrate that the proposed framework effectively integrates deep learning-based classification with object detection techniques to achieve accurate rice grain analysis. The system provides reliable grain detection, variety classification, defect identification, impurity detection, and automated quality grading, highlighting the potential of computer vision technologies for modern agricultural quality inspection systems.

6.2 FUTURE SCOPE

Although the proposed system achieves promising results in rice grain defect detection and quality grading, several improvements can further enhance its performance and real-world applicability. One possible improvement is the expansion of the dataset with a larger number of rice grain images captured under different lighting conditions, backgrounds, and grain orientations. A more diverse dataset would help improve the robustness and generalization capability of the trained models. Future work can also explore end-to-end deep learning architectures that perform grain detection and defect classification within a single model. This would simplify the pipeline and reduce computational complexity by eliminating the need for separate detection and classification stages. Another important direction is handling overlapping and occluded rice grains, which remains a challenging problem in dense grain images. Improving the model to accurately detect and separate overlapping grains would significantly enhance grading accuracy in real-world bulk samples. Additionally, the development of a real-time rice quality inspection system integrated with industrial conveyor belts or automated sorting machines can be explored. Such a system could continuously monitor rice samples during processing and provide immediate quality grading. The proposed model can also be extended to detect more types of defects and impurities, such as immature grains, foreign particles like stones or husk, making the system more comprehensive for practical rice quality evaluation. Finally, the system can be further enhanced as a mobile or web-based application, enabling farmers, traders, and quality inspectors to analyze rice samples using smartphone cameras or portable imaging devices, making automated rice grading more accessible in real-world agricultural environments.

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